UNIVERSITAS
GADJAH MADA

Kajian Komprehensif Topo Hidro WM – Penyusunan Peta Arahan Tindakan RG P Padang

Tim Sipil dan Tim Geospasial FT UGM

08 Agustus 2018, Jakarta, BRG

Hasil Kajian Komprehensif Topografi- Hidrologi – Hidrolika



UNDP – BRG – UGM

berawal dari istana….

setkab.go.id/terima-pokja-ugm-presiden-jokowi-minta-pemetaan-detil-lahan-gambut-di-tanah-air/

SEKRETARIAT KABINET REPUBLIK INDONESIA

BERANDA BERITA ARTIKEL PERATURAN INFORMASI G

Terima Pokja UGM, Presiden Jokowi Minta Pemetaan Detil Lahan Gambut Di Tanah Air

Oleh: **Humas** ; Diposkan pada: **3 Nov 2015 ; 12077 Views** Kategori: Berita

Presiden Joko Widodo (Jokowi) memandang serius masalah pengolahan lahan gambut ke depan. Untuk itu, Presiden meminta dilakukannya segera dilakukan pemetaan detil seluruh kawasan gambut di Indonesia, bukan kedalamannya tapi topografinya. Sehingga gerakan penyekatan kanal dan perbaikan restorasi air itu akan segera dilakukan dalam waktu beberapa bulan ke depan.



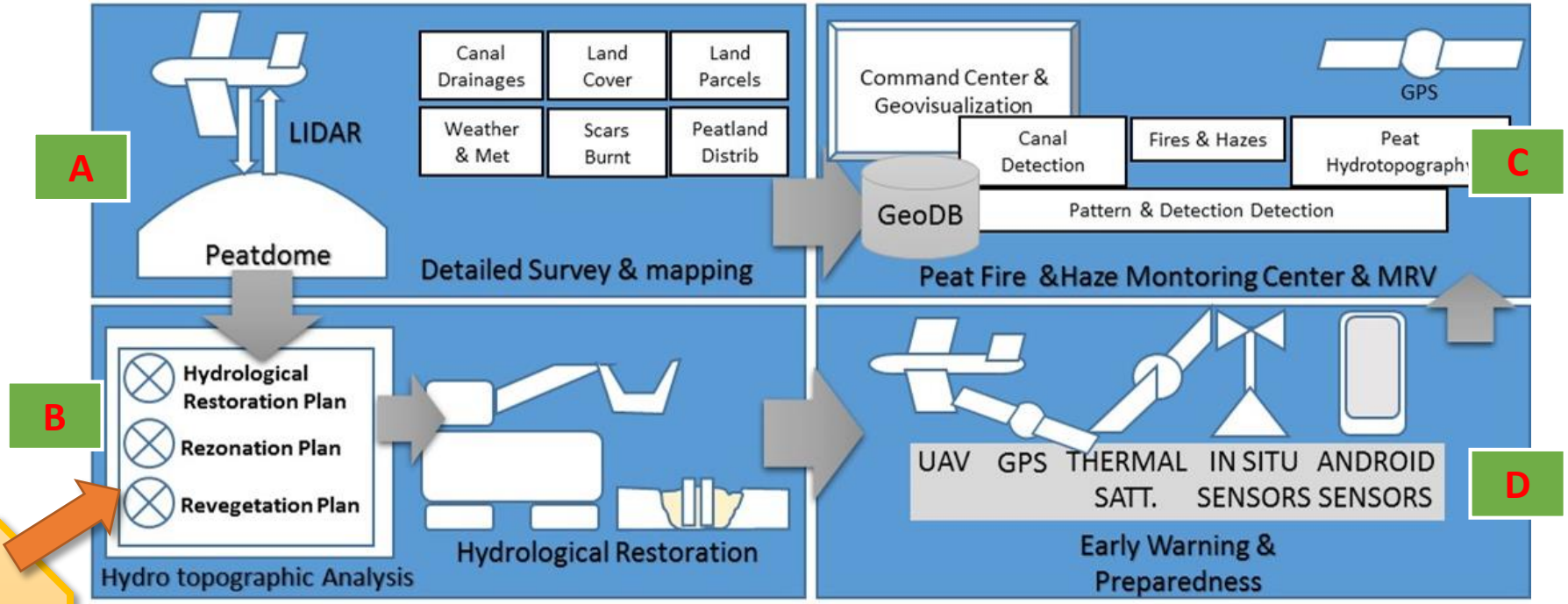
Presiden Jokowi didampingi sejumlah Menteri menerima Tim Pokja Karhutla dan Gambut, di Kantor Presiden, Jakarta, Selasa (3/11)

Permintaan tersebut disampaikan Presiden Jokowi saat menerima Kelompok Kerja Kebakaran Hutan dan Lahan dan Gambut (Pokja Karhutla dan Gambut) Universitas Gadjah Mada, di Istana Merdeka, Jakarta, Selasa (3/11) siang. Dalam kesempatan ini Pokja Karhutla dan Gambut UGM melaporkan hasil kajian yang dilakukan sejak tahun 1974 lalu.

"Dari hasil kajian, kami sampaikan bahwa resep untuk mengatasi atau mencegah kerusakan lahan gambut perlu adanya integrasi beberapa aspek, yaitu aspek rekayasa sosial, rekayasa teknis, dan juga aspek politik pembangunan ekonomi dalam pengelolaan HTI (Hutan Tanaman Industri) atau perkebunan. Ketiga aspek yang terintegrasi tadi perlu dibungkus dengan aspek legal serta tata ruang," kata Dwikorita, Rektor Universitas Gadjah Mada, seusai bersama Tim Pokja diterima Presiden Jokowi.

Ketua Pokja Karhutla UGM, Azwar Maas, yang merupakan ahli gambut mengatakan, gambut itu mempunyai kubah yang mempunyai simpanan air yang sangat besar. Ia menyebutkan, di kaki kubah itu ada 2 (dua) sungai, namun

Spatially referenced Data Utilization for Supporting Peatland Restoration



Posisi Peta Rencana Tindakan Restorasi Gambut Pulau Padang

Bagan Alir Penanganan Lahan Gambut Lestari



Sumber : Pak Djoko Luknanto

Data Input & Output Pemrosesan Survei dan Pemetaan Kondisi Hidrologis Lahan Gambut

UNIVERSITAS GADJAH MADA

Layer/Informasi Dasar

1. Kontur/Topografi
2. Ketebalan Gambut
3. Tutupan lahan (dari Foto)
4. Peta Saluran Alami
5. Peta Spasial Hujan Bulanan
6. Peta Jaringan Jalan
7. Peta Saluran Buatan
8. Peta Spasial Kimia Tanah (Konfirmasi)
9. Peta Batas Wilayah Administrasi
10. Peta Batas Status Penguasaan Lahan (APL, IUPH HTI)
11. Riwayat kebakaran

Produk Analisis

1. Peta Sub KHG
2. Neraca Air Spasial-Temporal
3. Simulasi Saluran (Air permukaan) : Eksisting (Tanpa Sekat)
4. Simulasi Saluran (Setelah ada tindakan dengan Sekat)

Dari hasil ini dihasilkan klasifikasi kerentanan/kebasahan

- Selisih muka air lahan dan saluran (kekritisan?)

Selanjutnya berdasarkan kriteria Pertanian dan Kehutanan (terasuk pertimbangka riwayat kebakaran?) dihasilkan :

- Rencana Tindakan

Akuisisi Data - LIDAR

PROJECT DESIGN & PLANNING
- Recommendation for SC
- Flight Permitt (SC, SO)
- Existing Benchmark and Its Coordinate
- Flight Plan
- Logistic and Fuel Preparation

Tide Observation and Adjustment → **Ground Control Establishment**

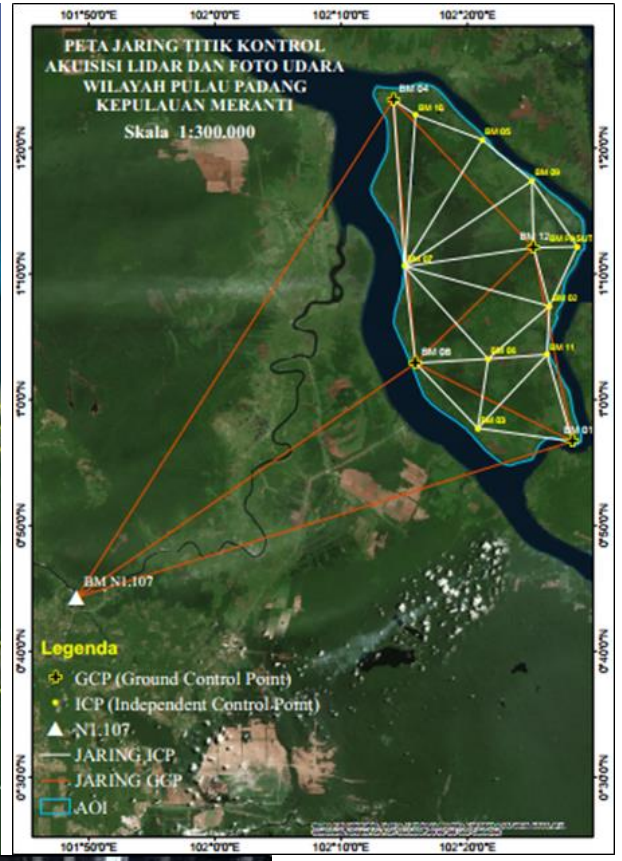
Aerial Data Acquisition
- Lidar
- Digital Stereo Aerial Photo

Ground Kinematic Observation

Preliminary Processing
- GPS/INS trajectory
- Lidar Signal Processing and World Transformation
- Aerial Photo Processing

Final Processing
- Point Cloud Classification and Editing
- Contour Generation
- Orthophoto Processing
- Topographic Feature Extraction
- Cartographic Processing and Editing









Akuisisi Data LIDAR

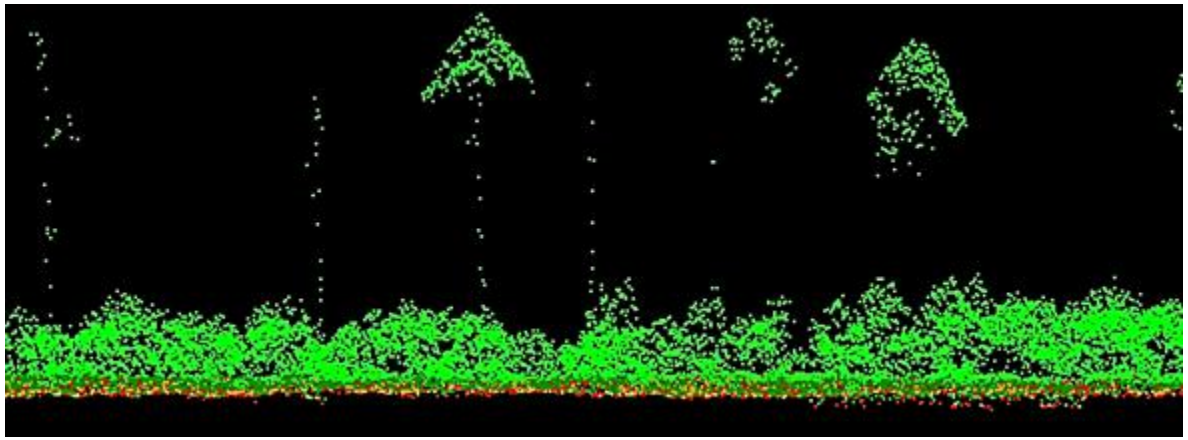


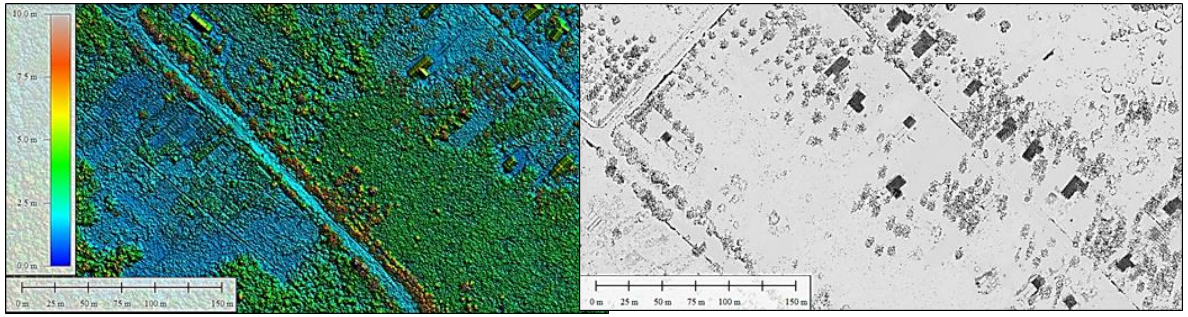


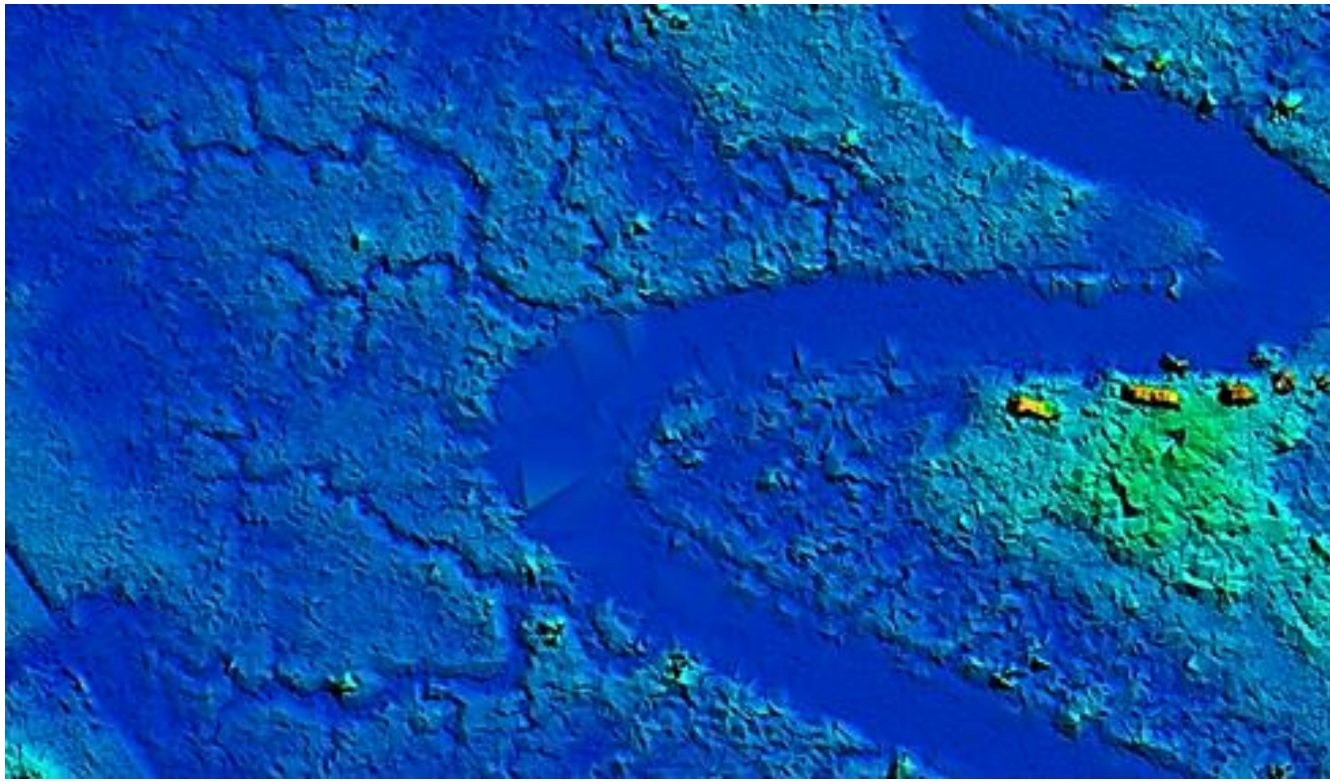


Pilatus PC6	RCD30 Digital Camera	ALS70 Laser Scanner & Controller
Camera Hole	Photogrammetric Camera	Up to 500 KHz Pulse
3 unit Pilatus PC6	Pixel size 8956 x 6708	FOV 0° - 75°
Turboprop Engine	Multispectral RGBN Imagery	Full Waveform
Up to 10.000 ft	2D Mechanical Motion Compensation	Unlimited range returns from each outbond pulse in greater detail
Capacity up to 10 passengers		AutoScan : automatically adjust scan rate in speed aircraft varies
		Double the scan rate

Data Masukan





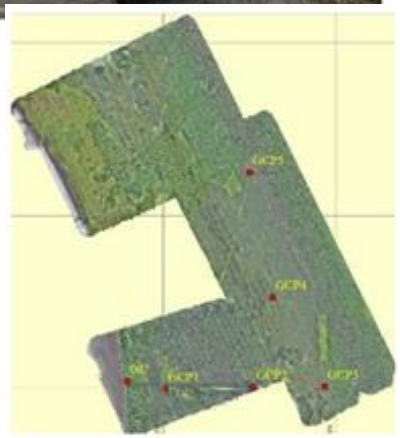


Akusisi Data UAV

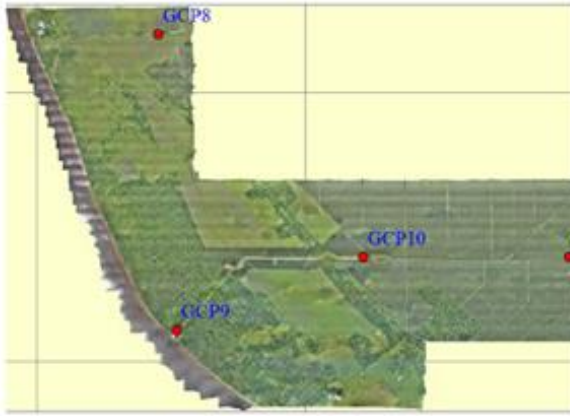




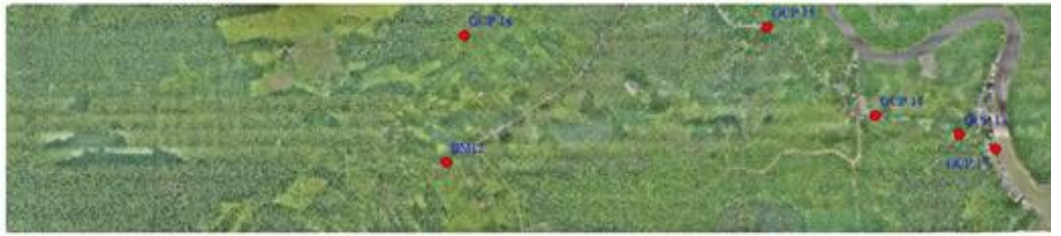




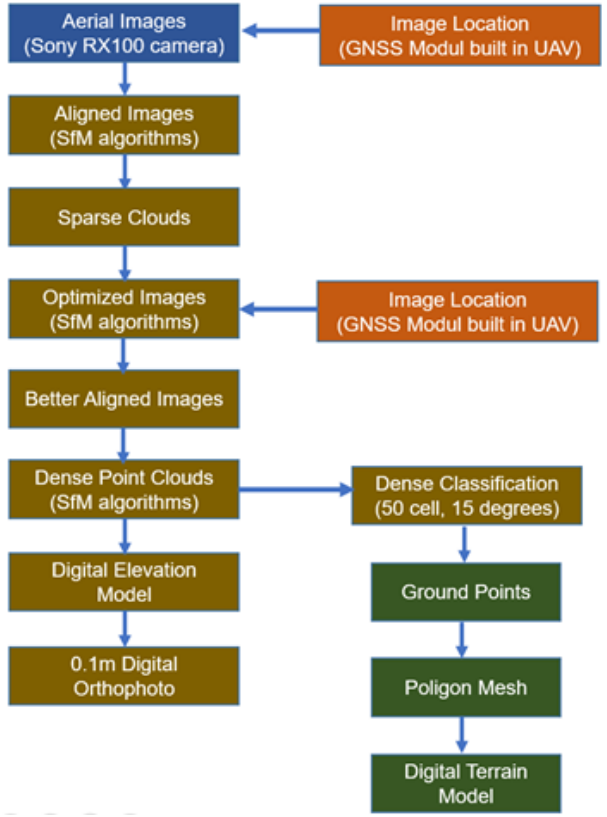
zona a



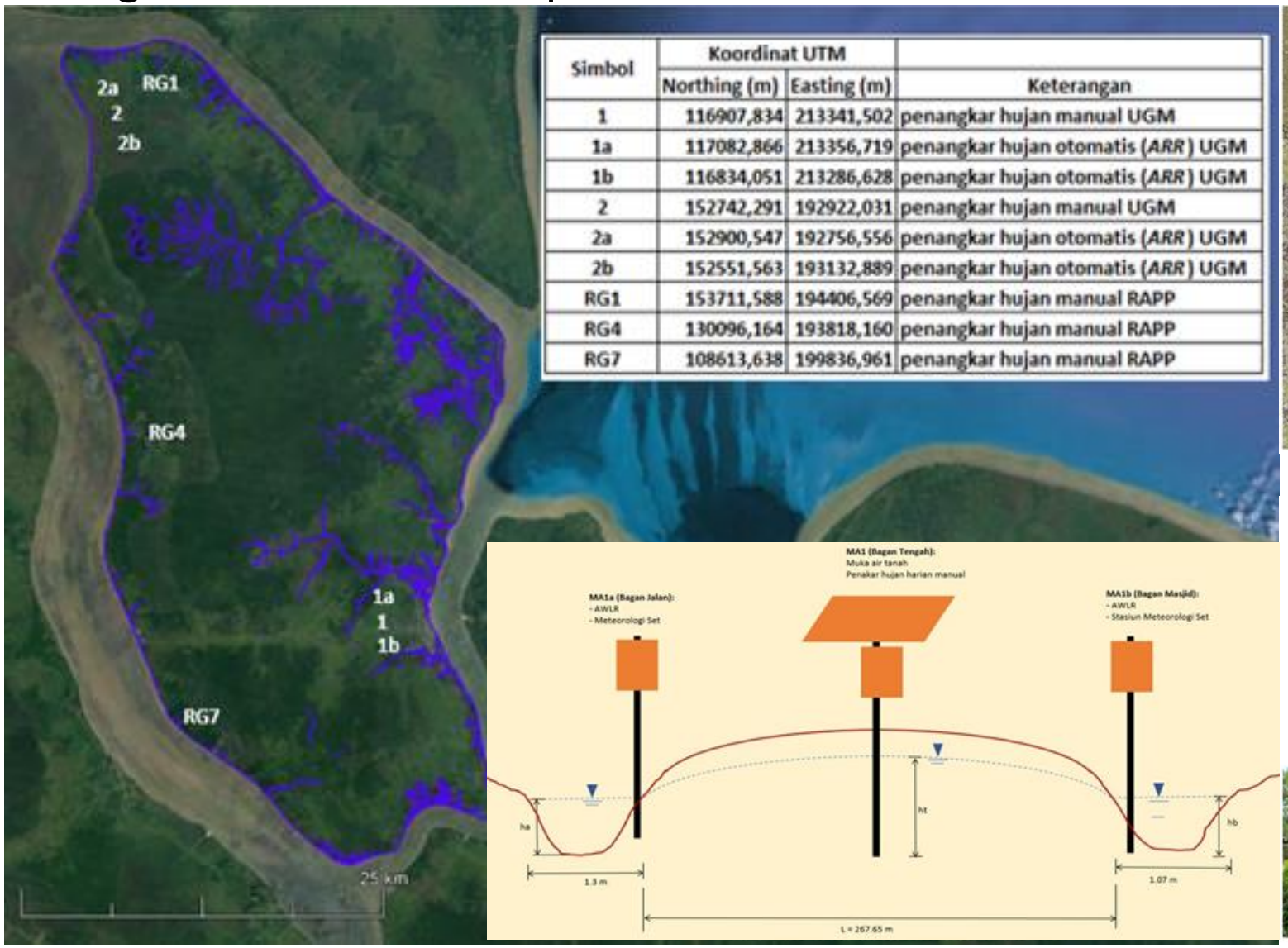
zona b



zona c

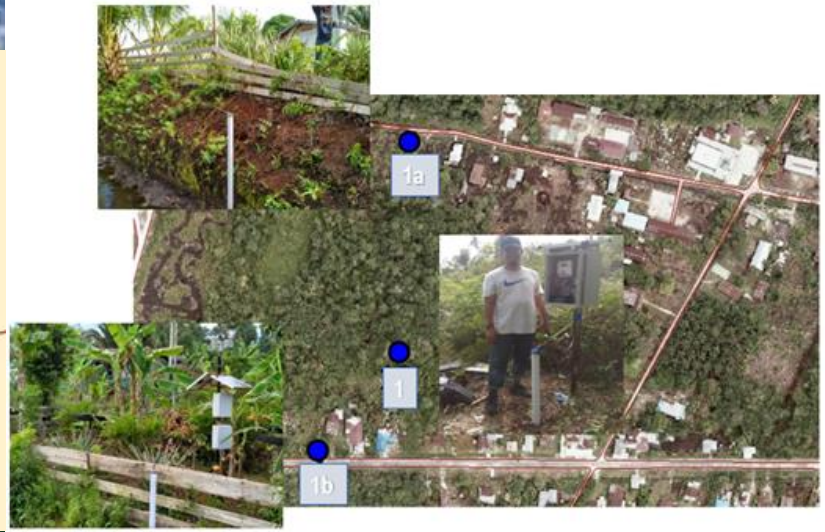


Pengukuran Curah Hujan

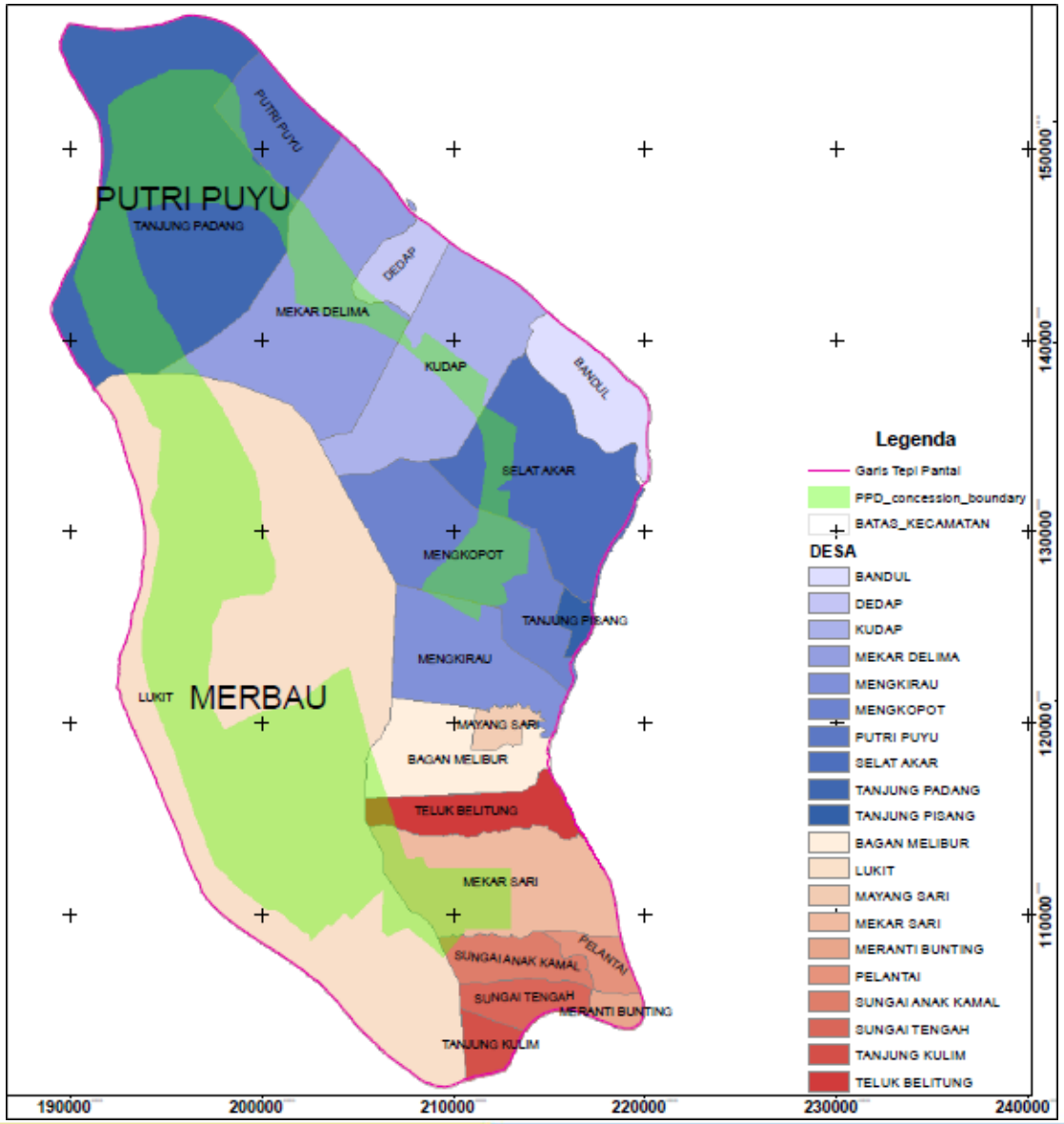


Simbol	Koordinat UTM		Keterangan
	Northing (m)	Easting (m)	
1	116907,834	213341,502	penangkar hujan manual UGM
1a	117082,866	213356,719	penangkar hujan otomatis (*ARR*) UGM
1b	116834,051	213286,628	penangkar hujan otomatis (*ARR*) UGM
2	152742,291	192922,031	penangkar hujan manual UGM
2a	152900,547	192756,556	penangkar hujan otomatis (*ARR*) UGM
2b	152551,563	193132,889	penangkar hujan otomatis (*ARR*) UGM
RG1	153711,588	194406,569	penangkar hujan manual RAPP
RG4	130096,164	193818,160	penangkar hujan manual RAPP
RG7	108613,638	199836,961	penangkar hujan manual RAPP



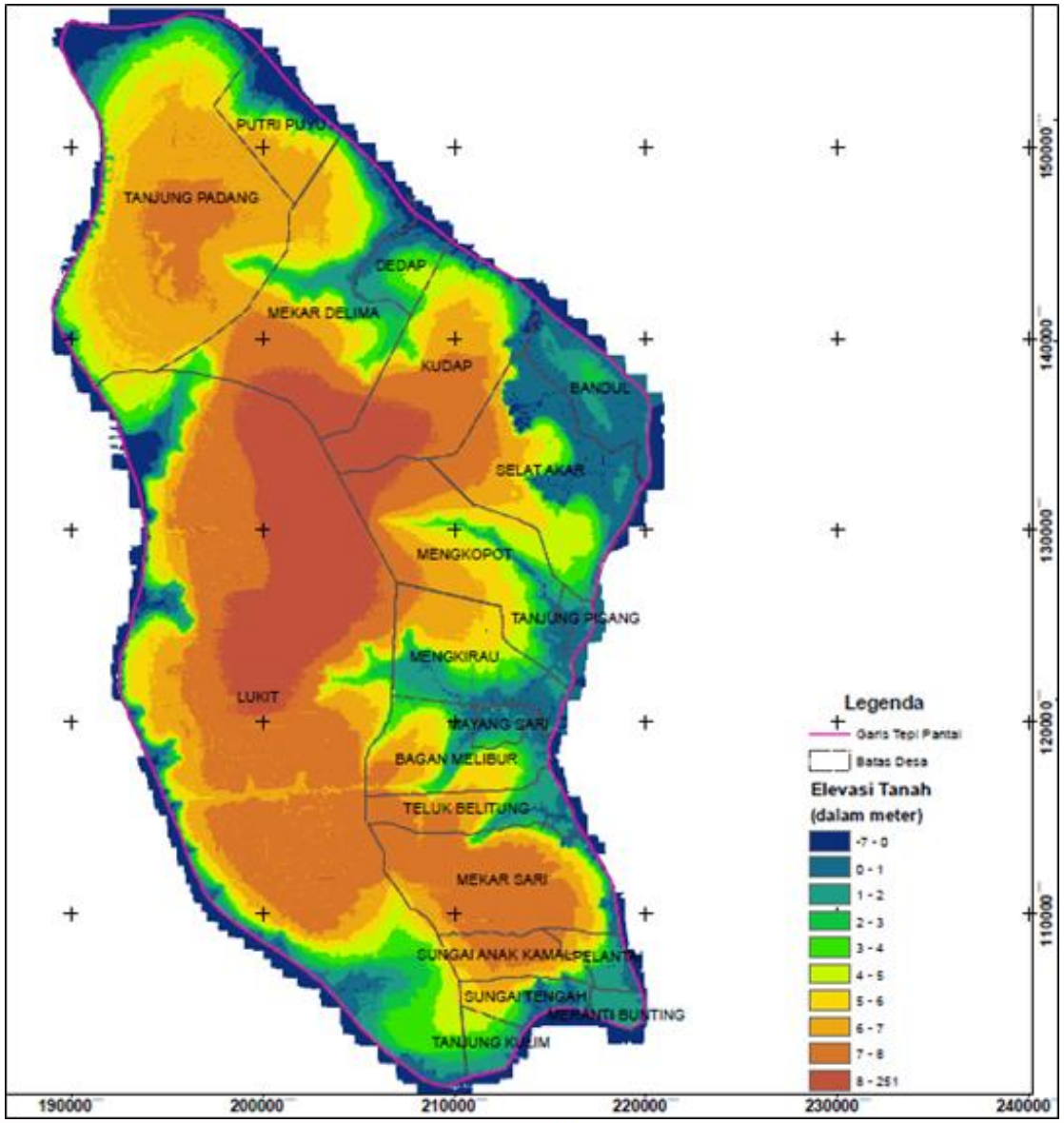


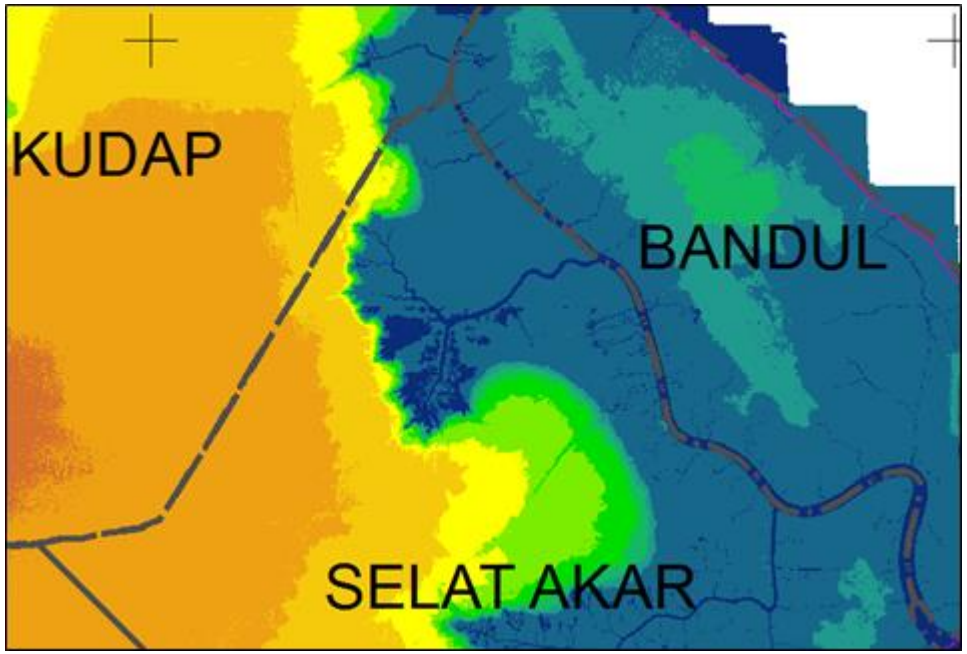
Data Masukan - WIlayah Administrasi & Konsesi

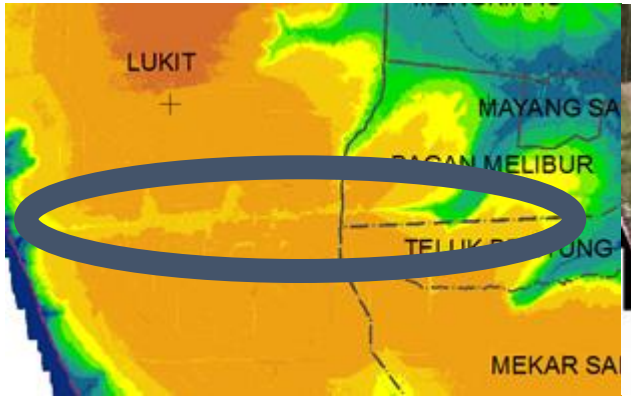


Kecamatan Merbau	Kecamatan Tasik Teluk Puyu
1. Kelurahan Teluk Belitung	1. Desa Bandul
2. Desa Bagan Melibur	2. Desa Dedap
3. Desa Lukit	3. Desa Kudap
4. Desa Mekar Sari	4. Desa Mekar Delima
5. Desa Meranti Bunting	5. Desa Mengkirau
6. Desa Pelantai	6. Desa Mengkopot
7. Desa Tanjung Kulim	7. Desa Putri Puyu
8. Desa Mayang Sari	8. Desa Selat Akar
9. Desa Sungai Anak Kamal	9. Desa Tanjung Padang
	10. Desa Tanjung Pisang

Data Masukan - Topografi

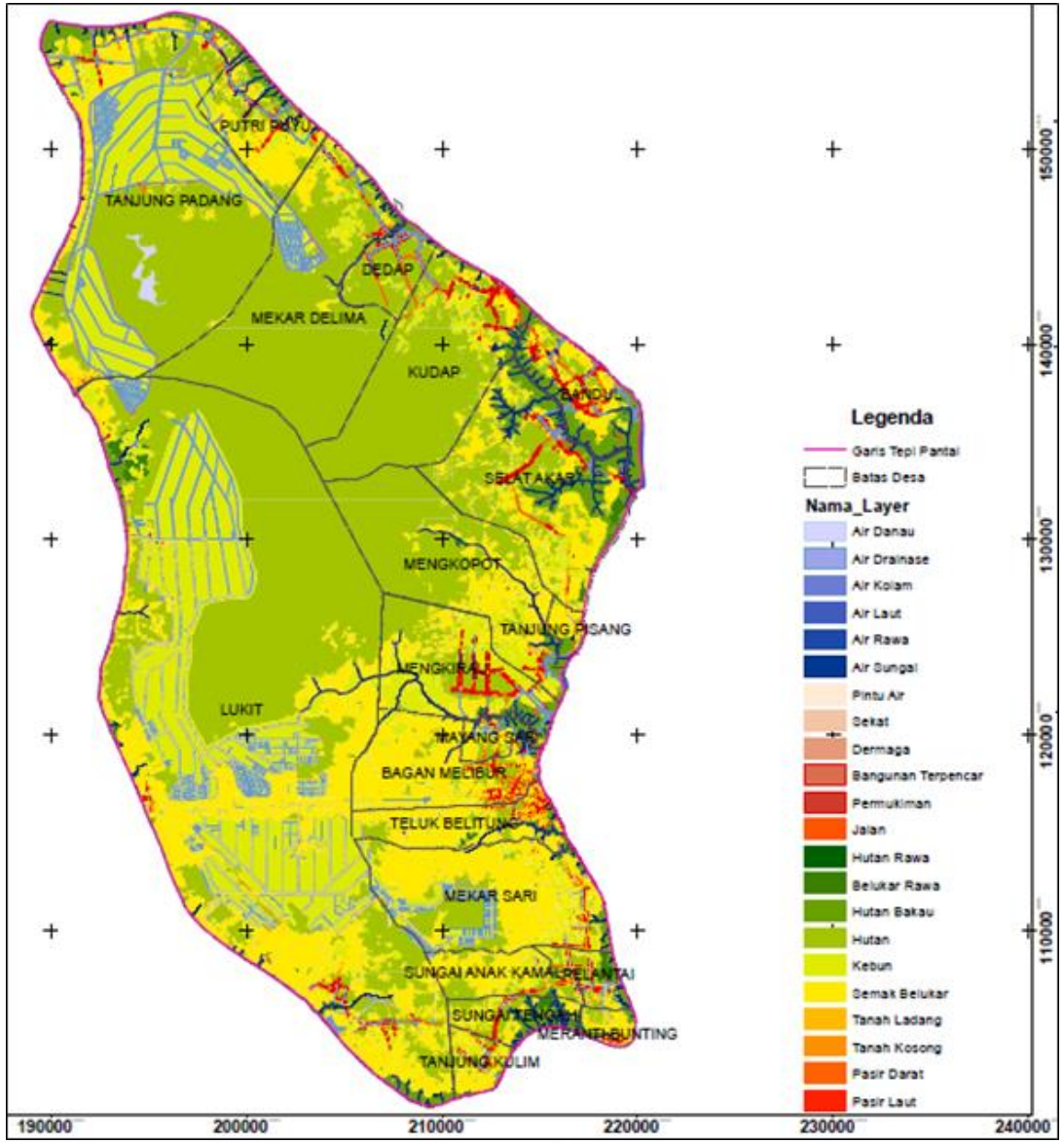








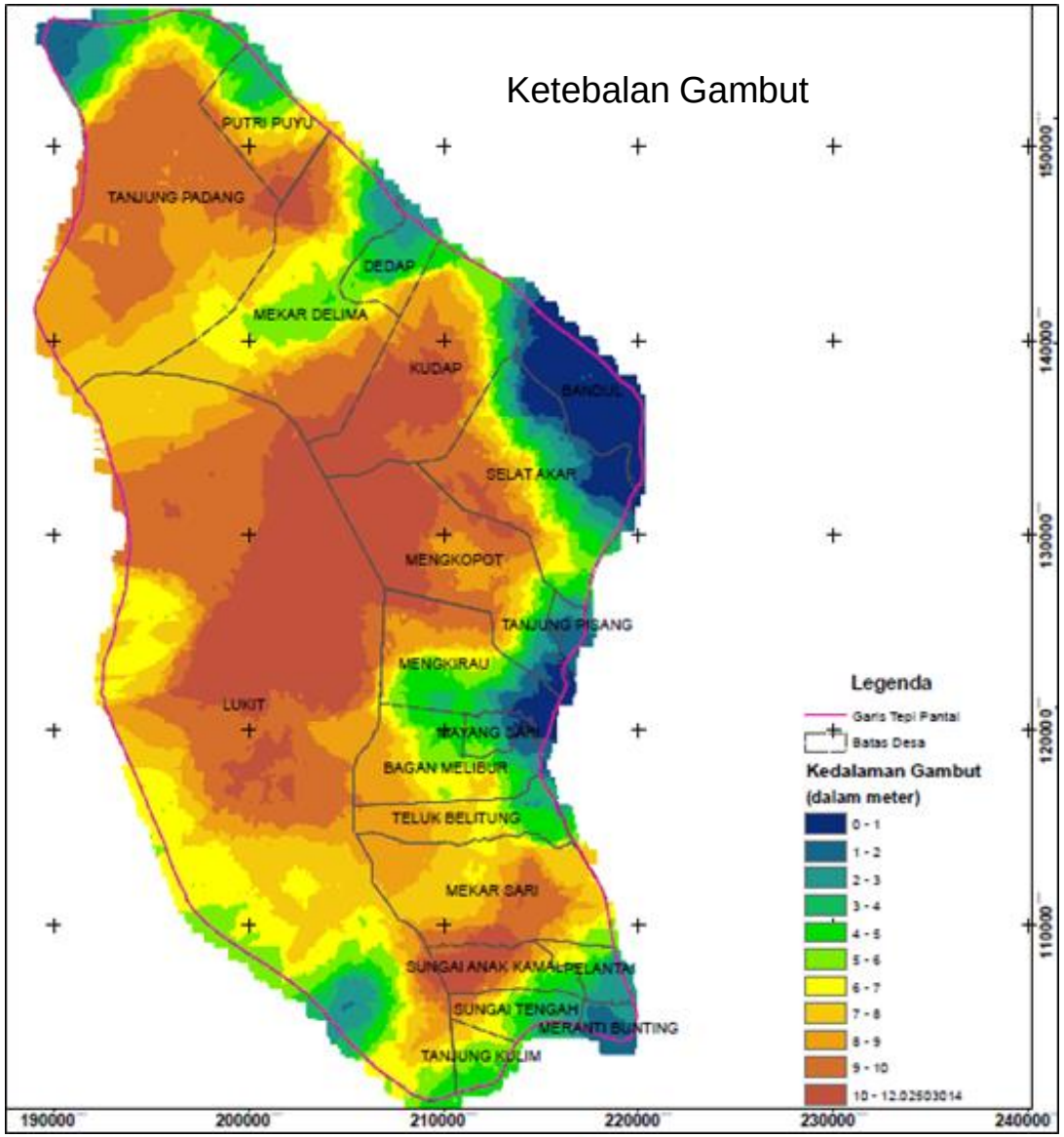
Data Masukan - Tutupan Lahan



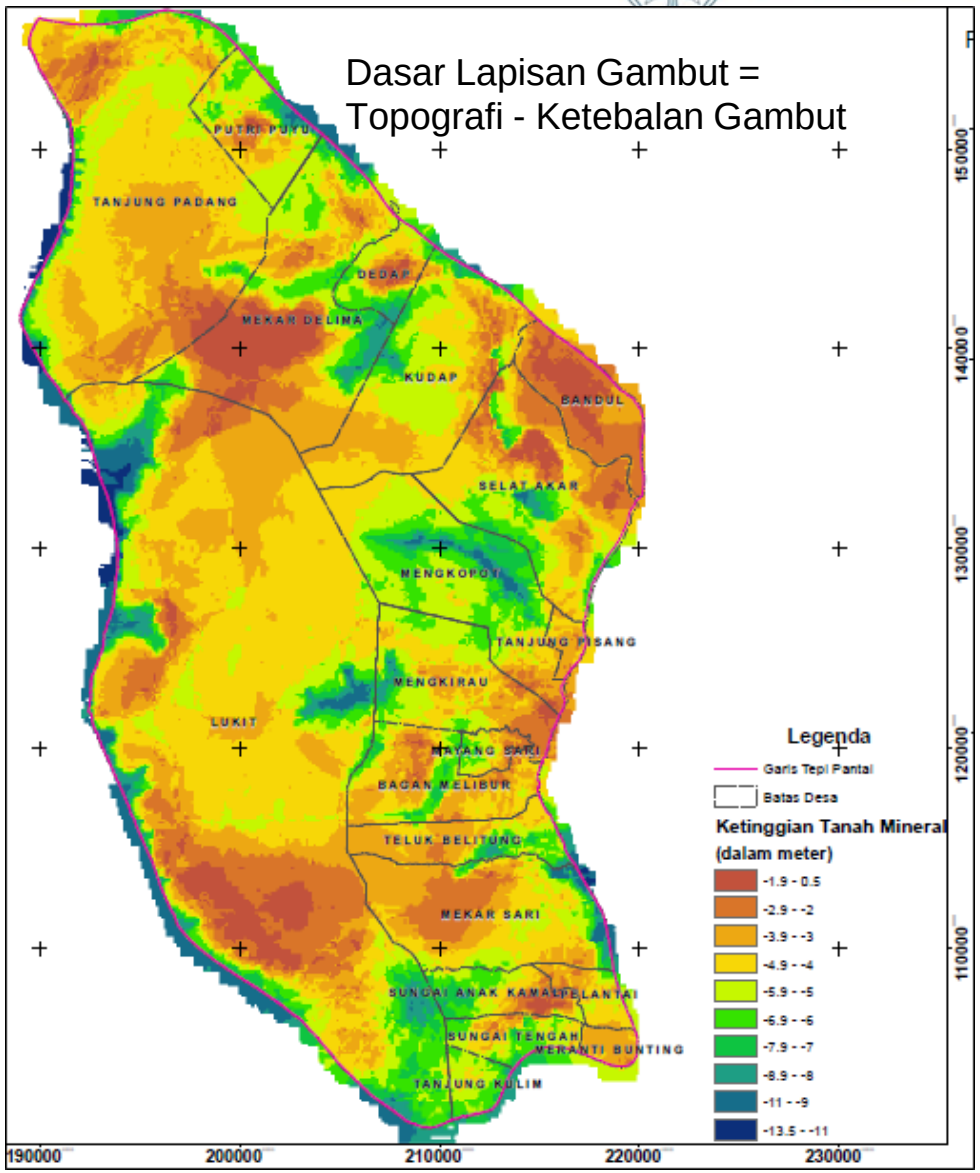
No	Kelas Penutup Lahan	Luas (Hektar)	Luas (%)
1	Hutan	49,906.70	45.01%
2	Semak Belukar	26,608.98	24.00%
3	Kebun	24,075.55	21.71%
4	Hutan Bakau	5,932.49	5.35%
5	Belukar Rawa	1,343.53	1.21%
6	Permukiman	1,327.68	1.20%
7	Jalan	396.64	0.36%
8	Air Drainase	391.20	0.35%
9	Air Sungai	308.54	0.28%
10	Air Danau	198.65	0.18%
11	Bangunan Terpencar	167.99	0.15%
12	Tanah Ladang	105.76	0.10%
13	Tanah Kosong	48.53	0.04%
14	Pasir Laut	40.43	0.04%
15	Hutan Rawa	19.14	0.02%
16	Dermaga	5.73	0.01%
17	Air Kolam	5.18	0.00%
18	Air Rawa	3.70	0.00%
19	Pasir Darat	0.86	0.00%
20	Sekat	0.35	0.00%
21	Pintu Air	0.02	0.00%
Luas Total KHG Pulau Padang		**110,887.66**	**100%**

Data Masukan - Ketebalan Gambut

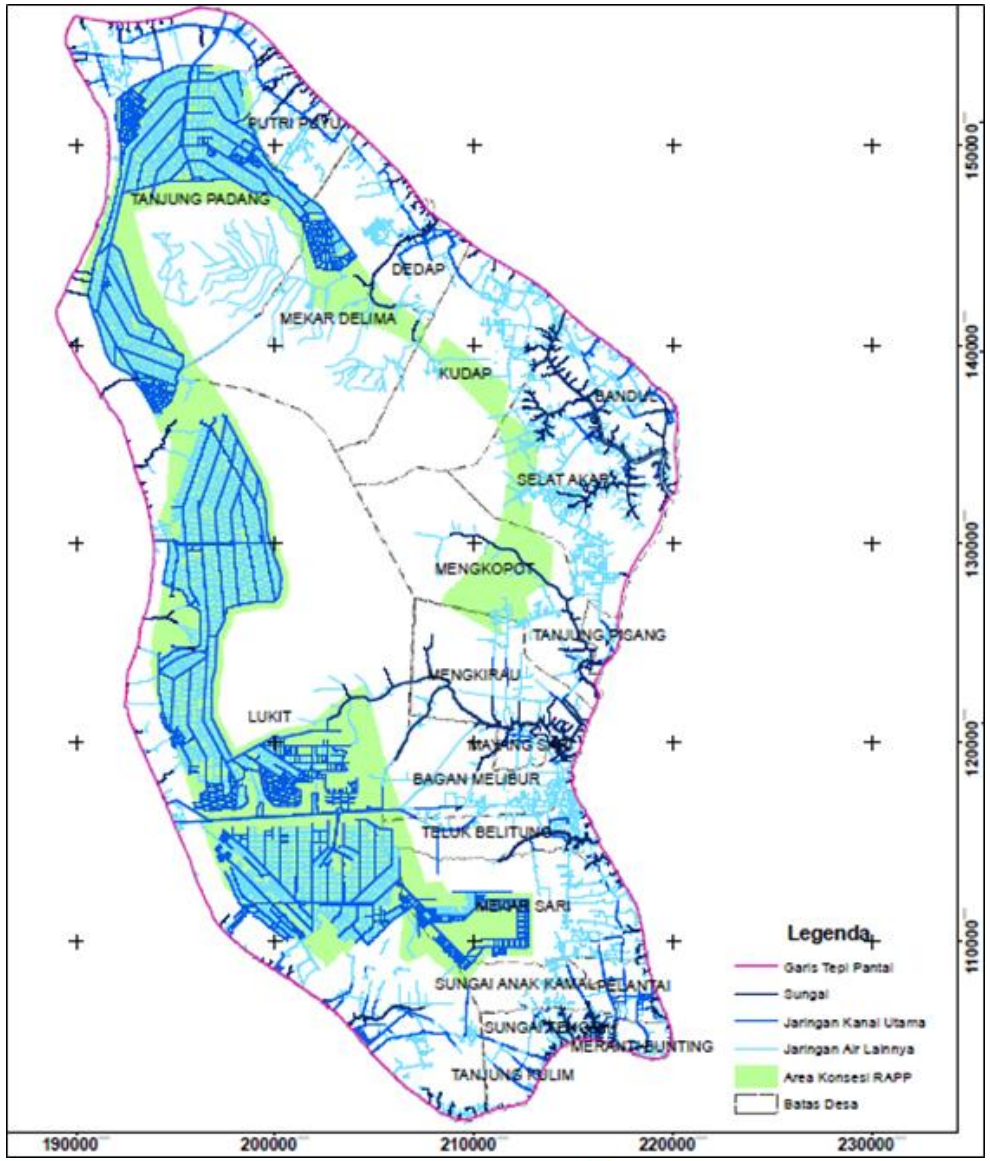
Ketebalan Gambut

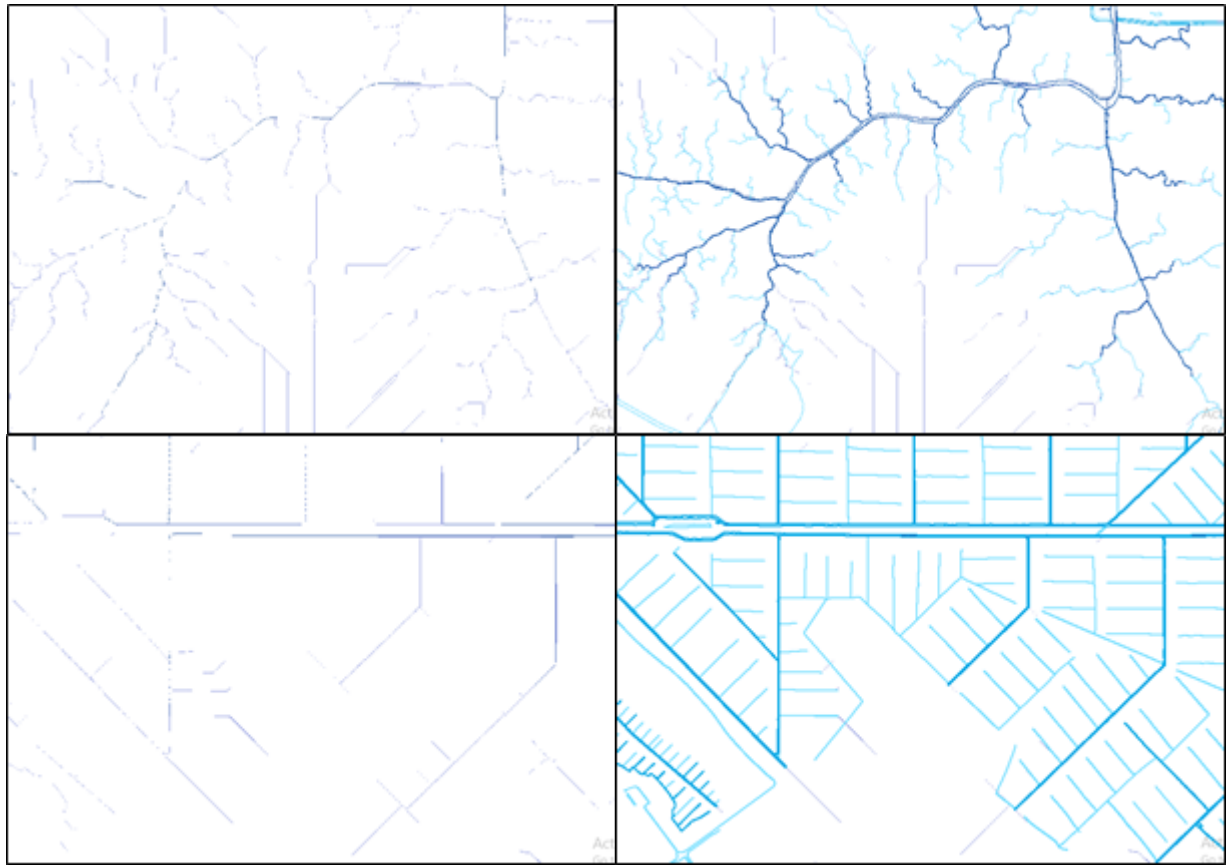


Dasar Lapisan Gambut =
Topografi - Ketebalan Gambut



Data Masukan - Saluran Alami dan Buatan

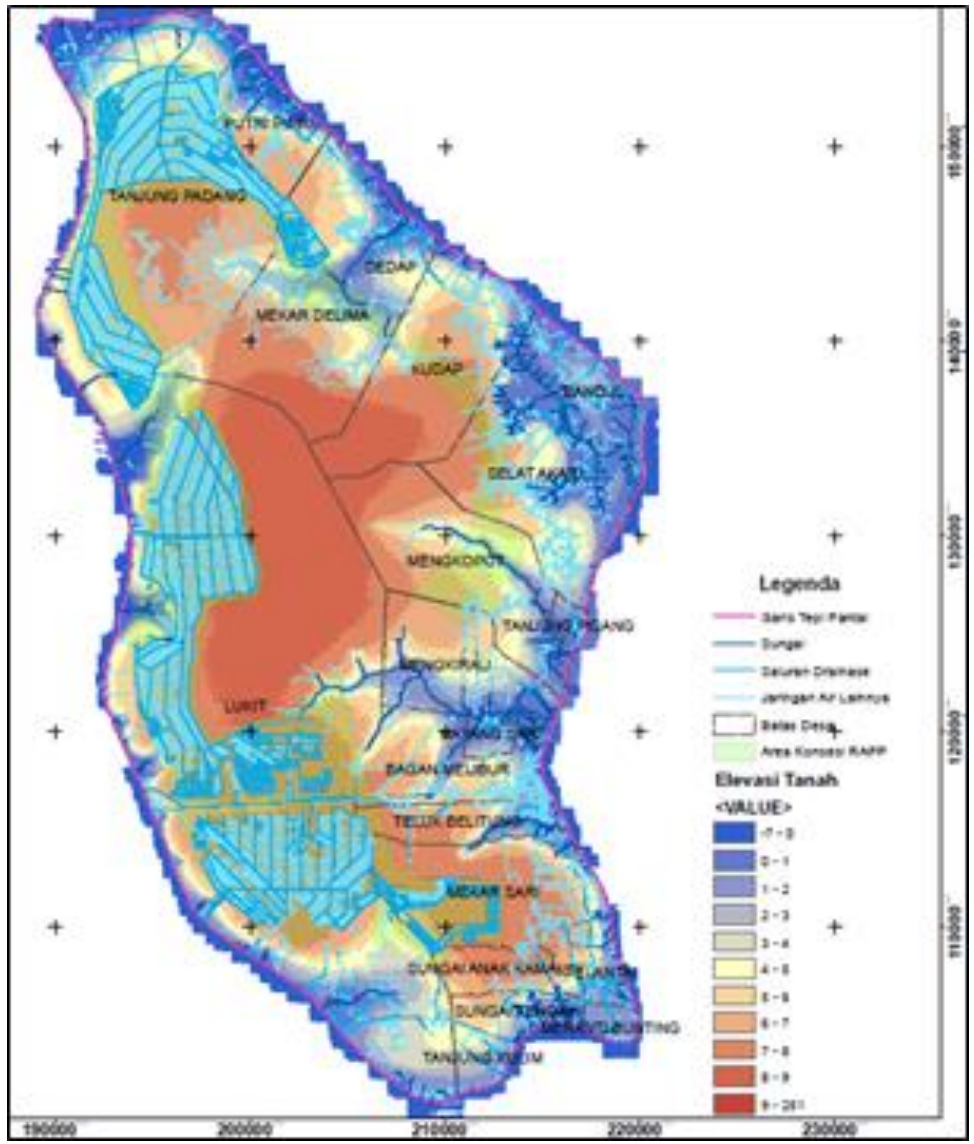


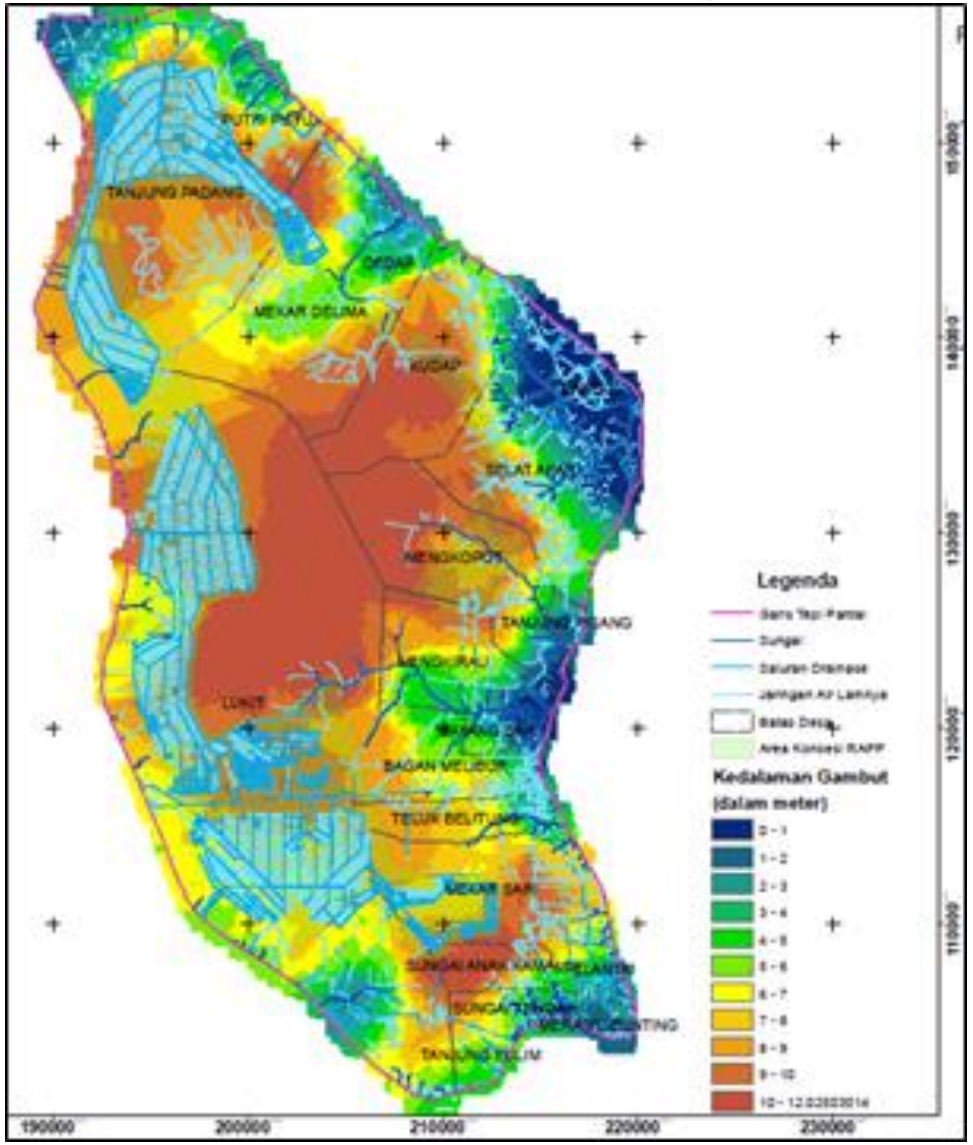


Hasil pemodelan aliran

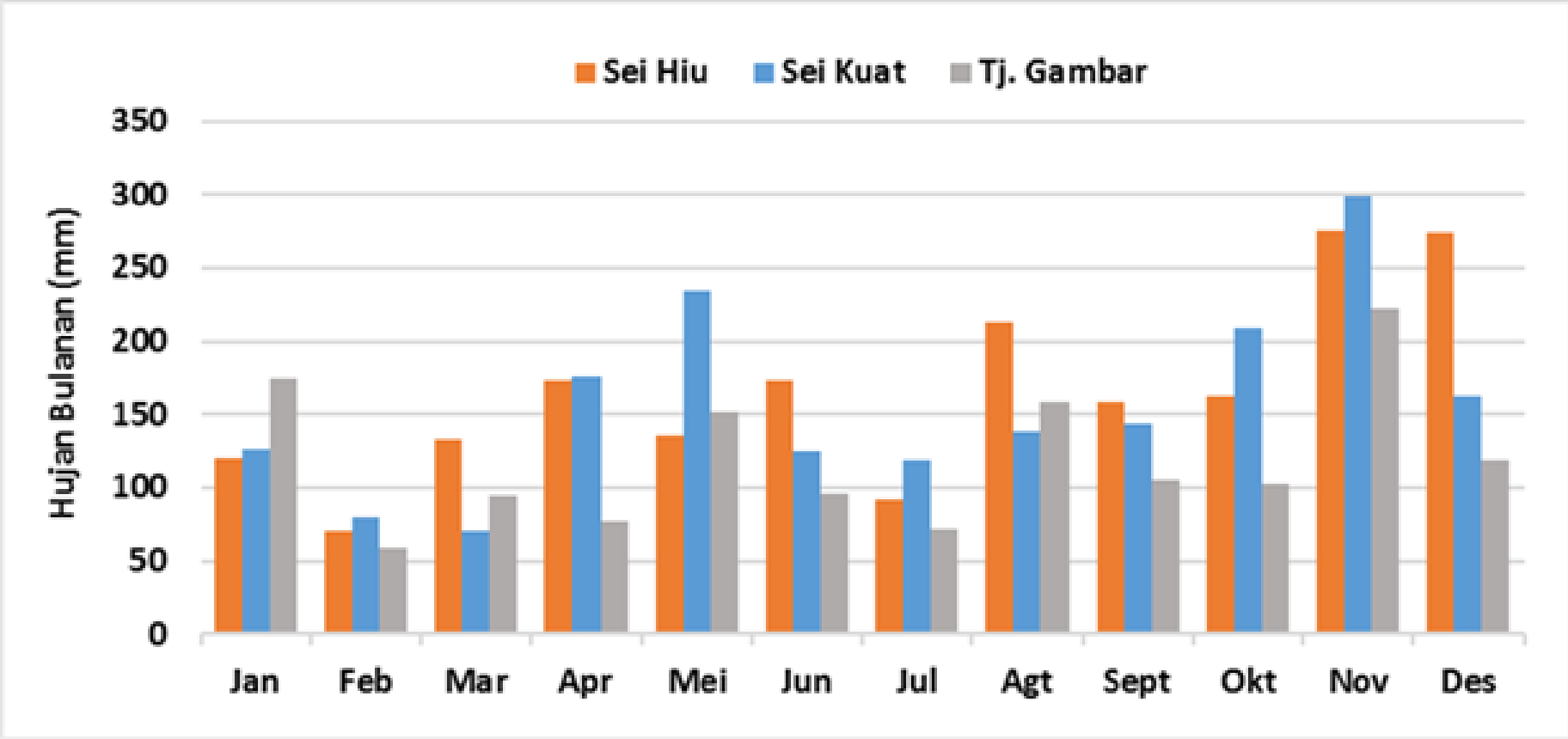
Digitasi saluran

Data Masukan - Layout Jaringan (Topo vs Ketebalan)



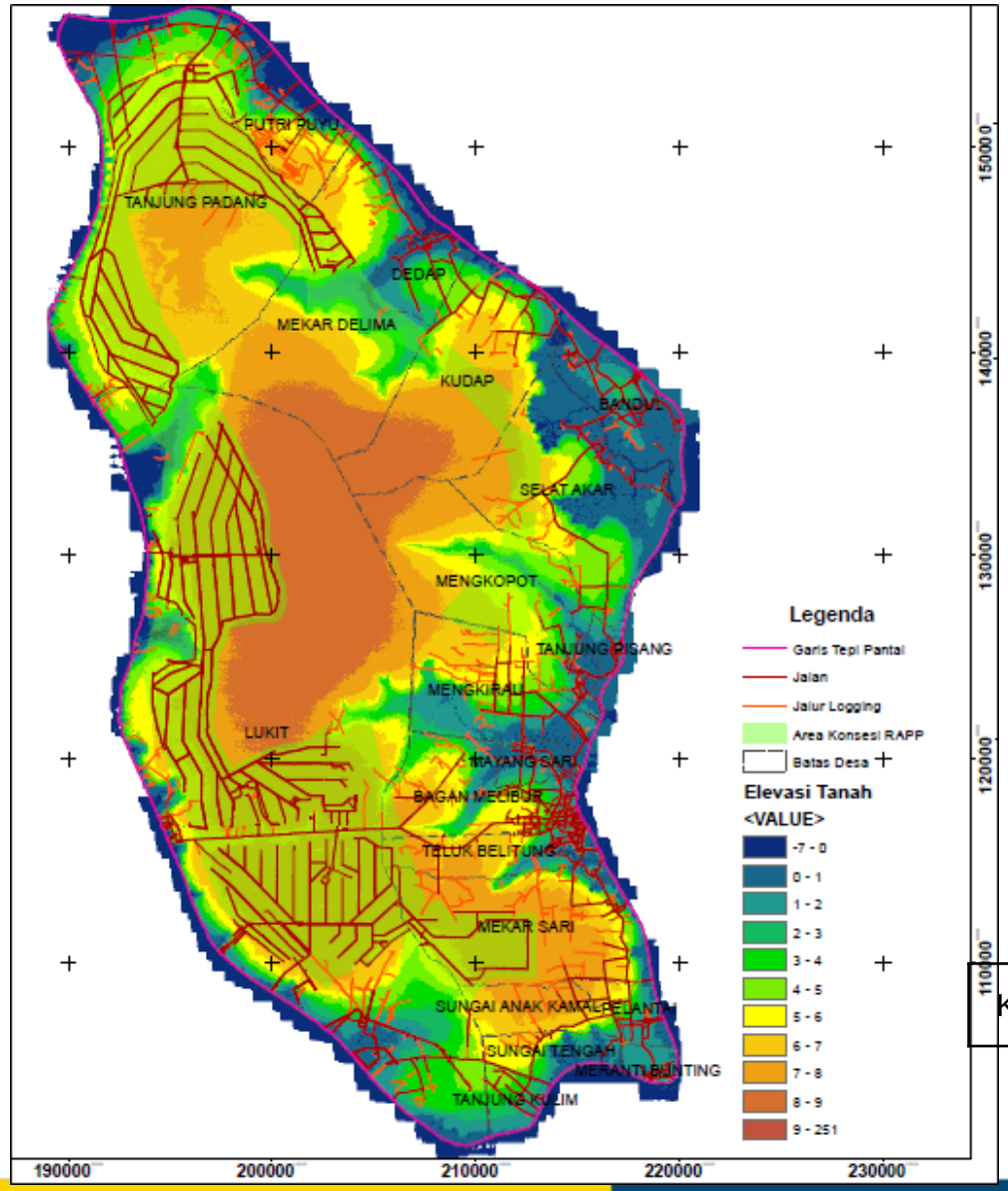


Data Masukan - Hidroklimatologi

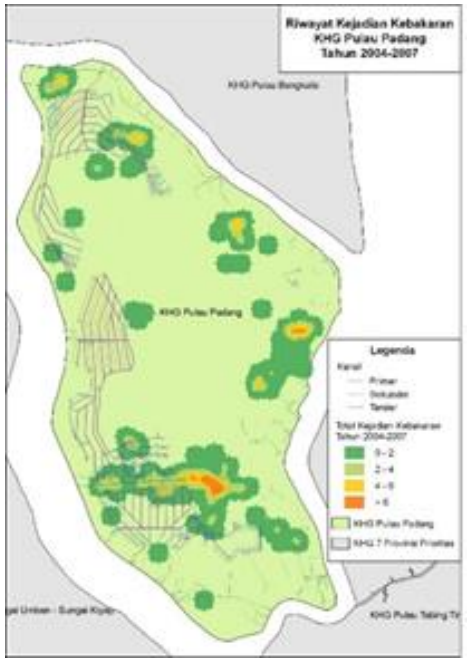


Hari tanpa hujan berturutan dalam sebulan berkisar antara 10 hari – 25 hari, dengan puncak bulan kering pada Februari dan Juli. Berdasarkan olahan data evapotranspirasi dari satelit GLDAS NOAH025 Monthly v2.1 tahun 2001 – 2016 diperoleh nilai evapotranspirasi bulanan 100 mm – 125 mm

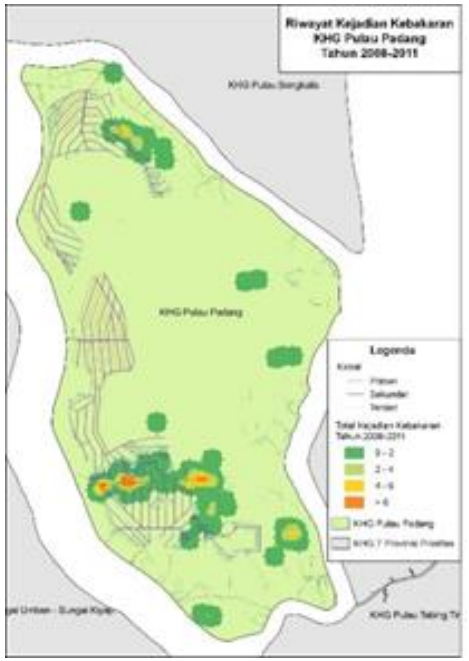
Data Masukan - Jaringan Jalan & Jalur Logging



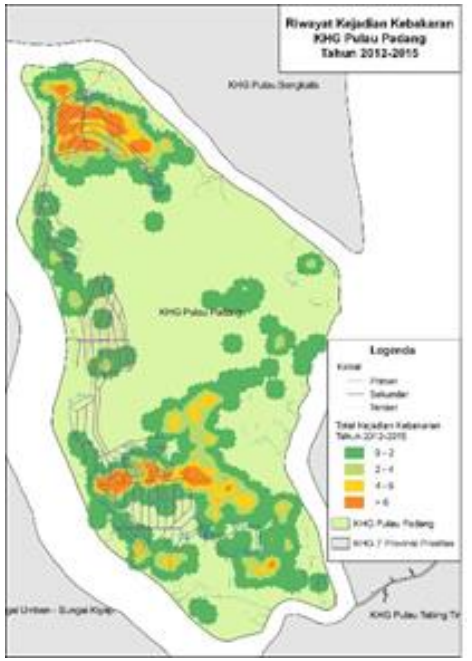
Data Masukan - Riwayat Kebakaran



Kejadian Kebakaran 2004-2007

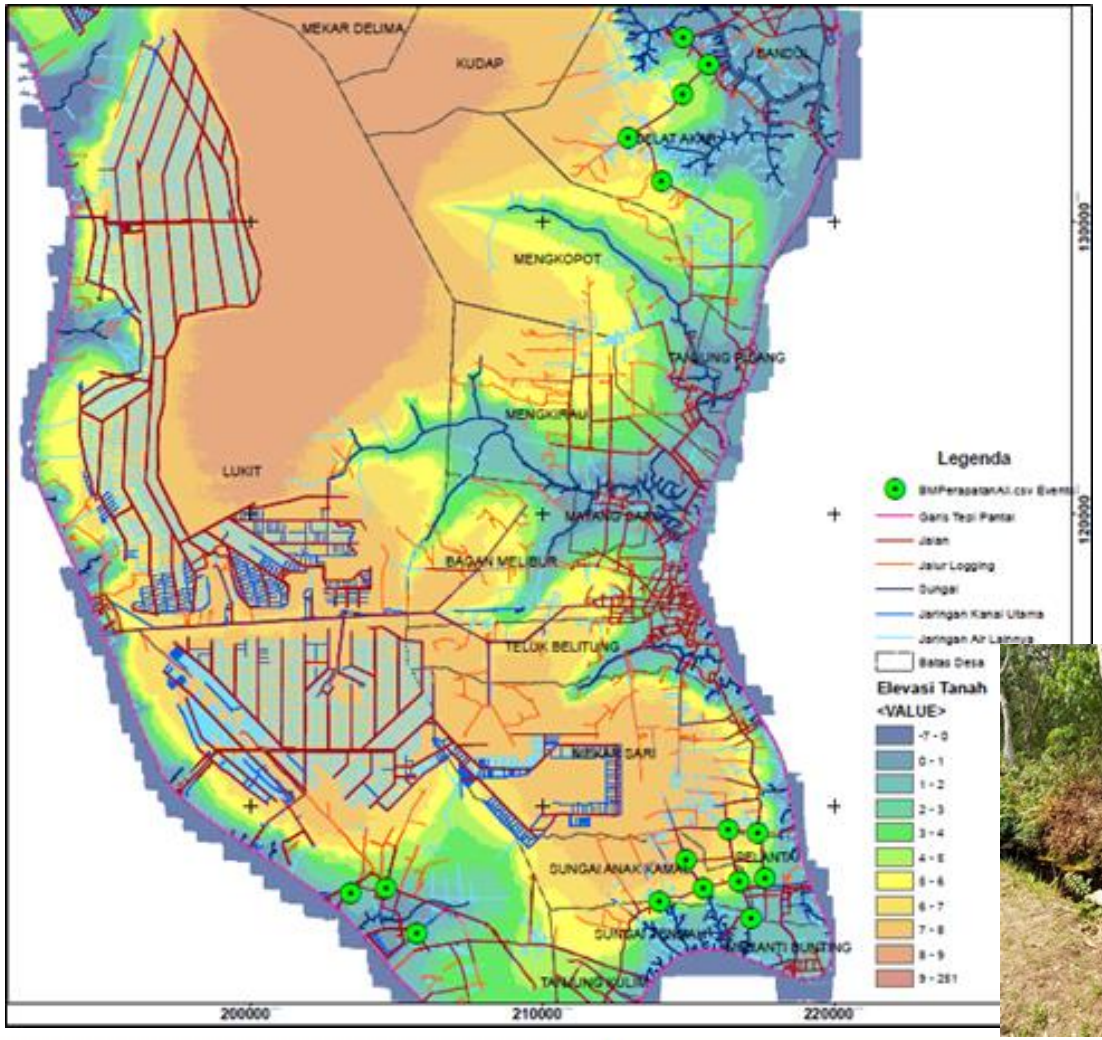


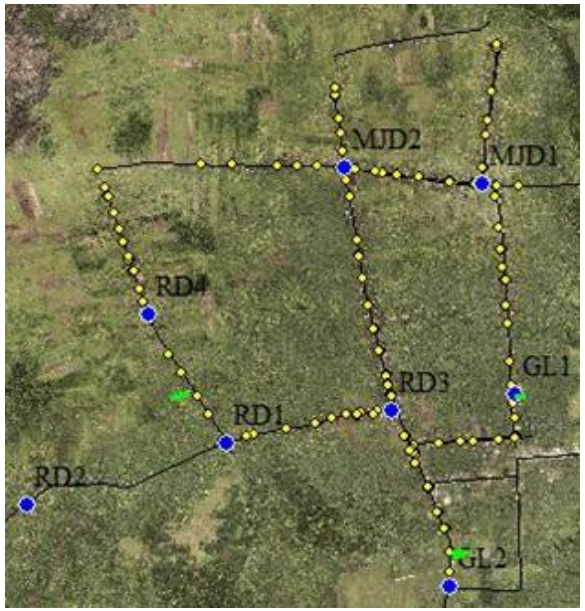
Kejadian Kebakaran 2008-2011

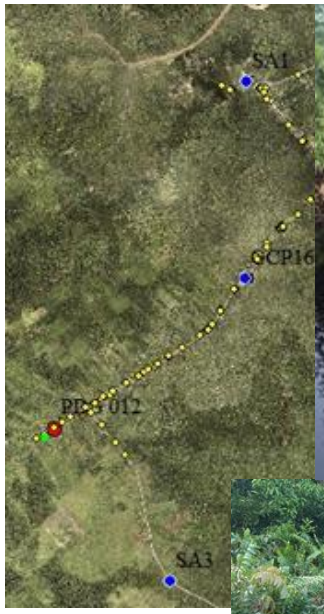


Kejadian Kebakaran 2012-2015

Evaluasi Akurasi











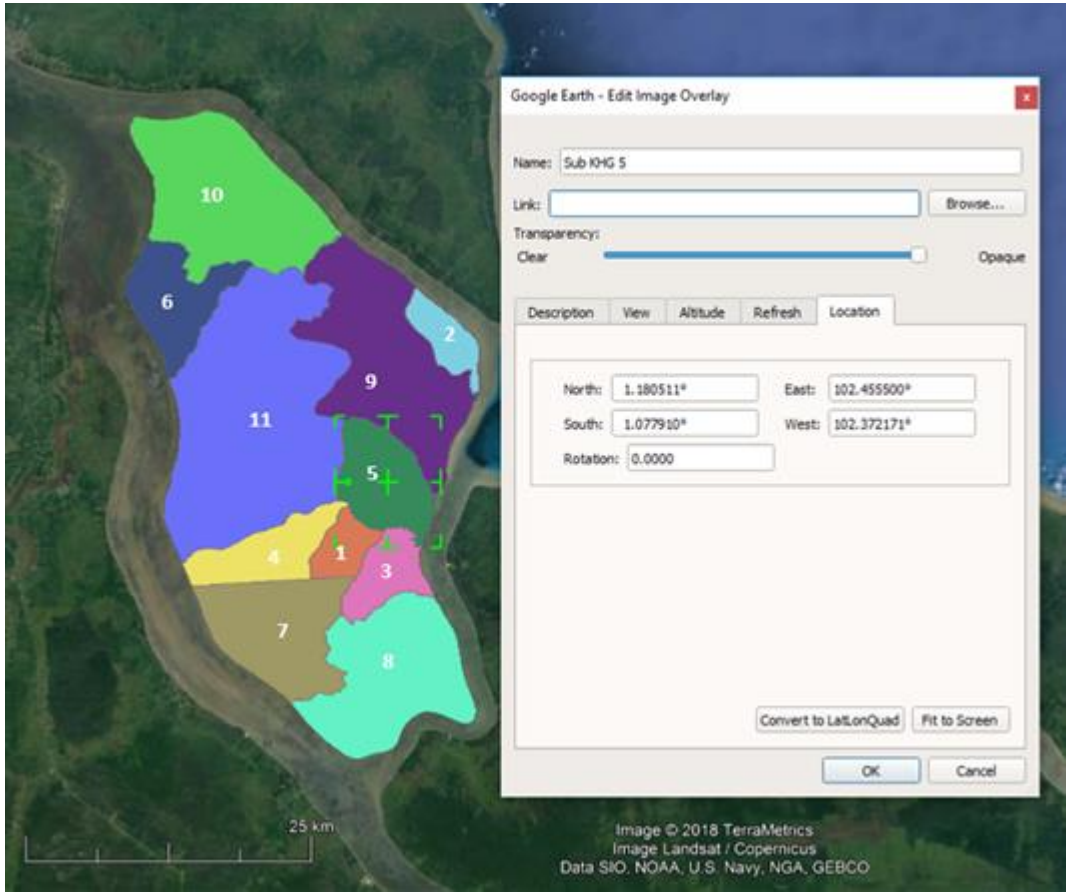


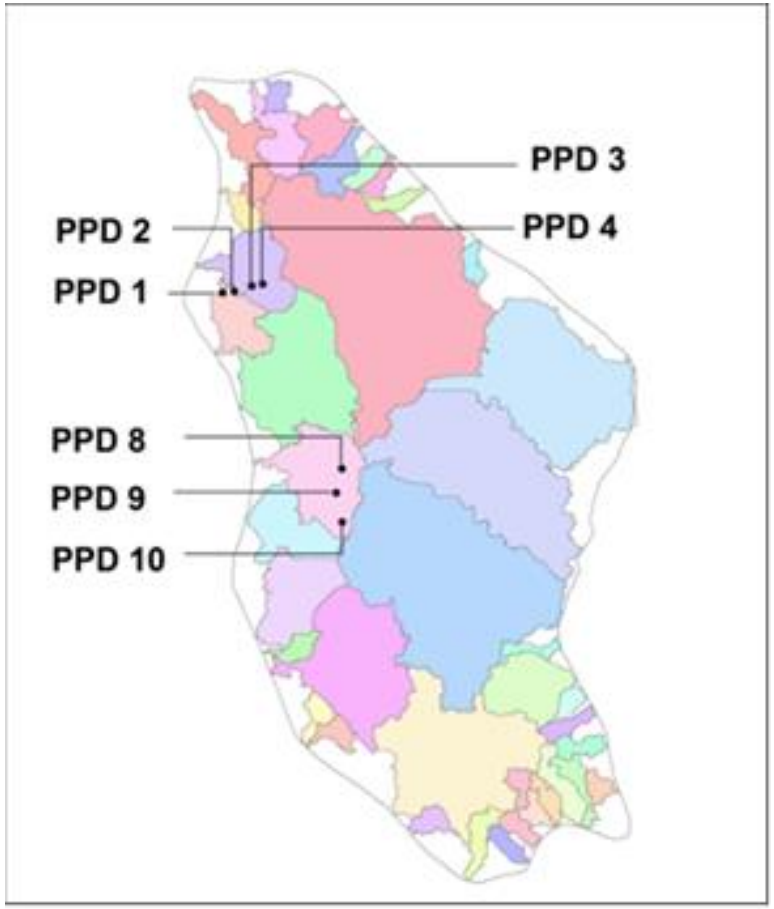




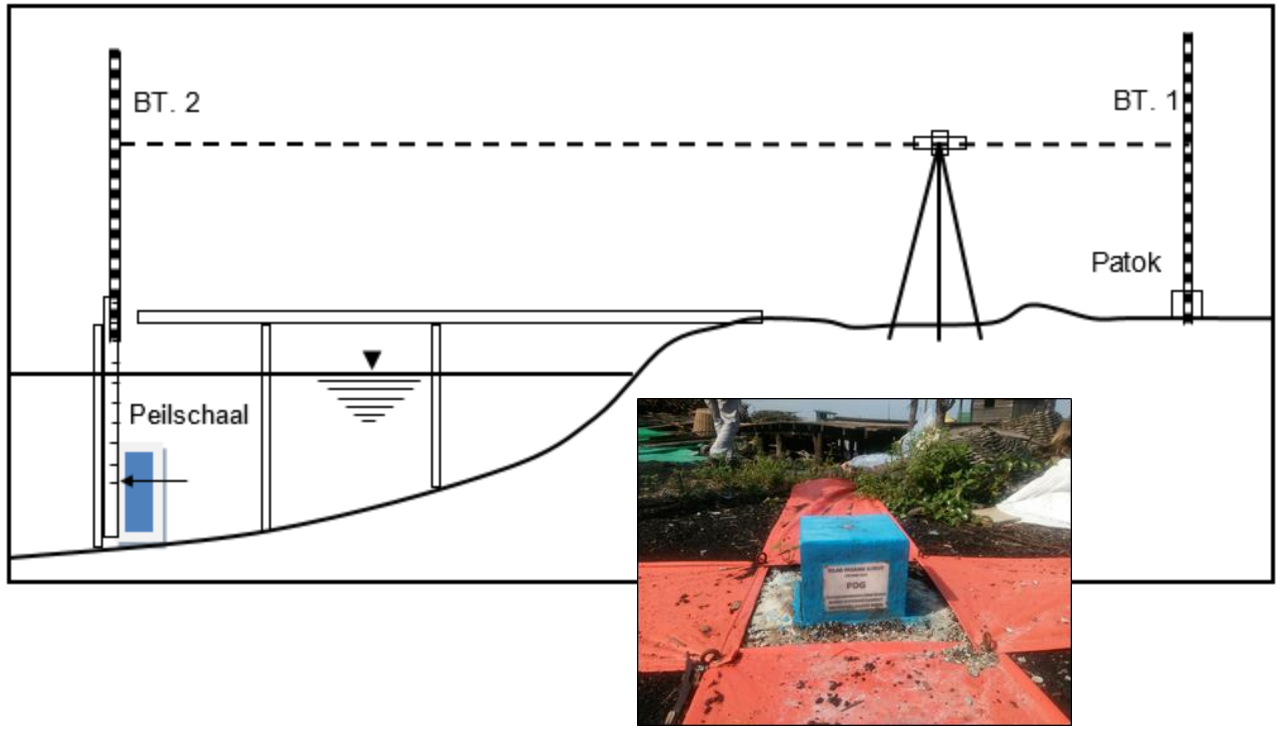


Pengukuran Curah Hujan Per Area Sub KHG



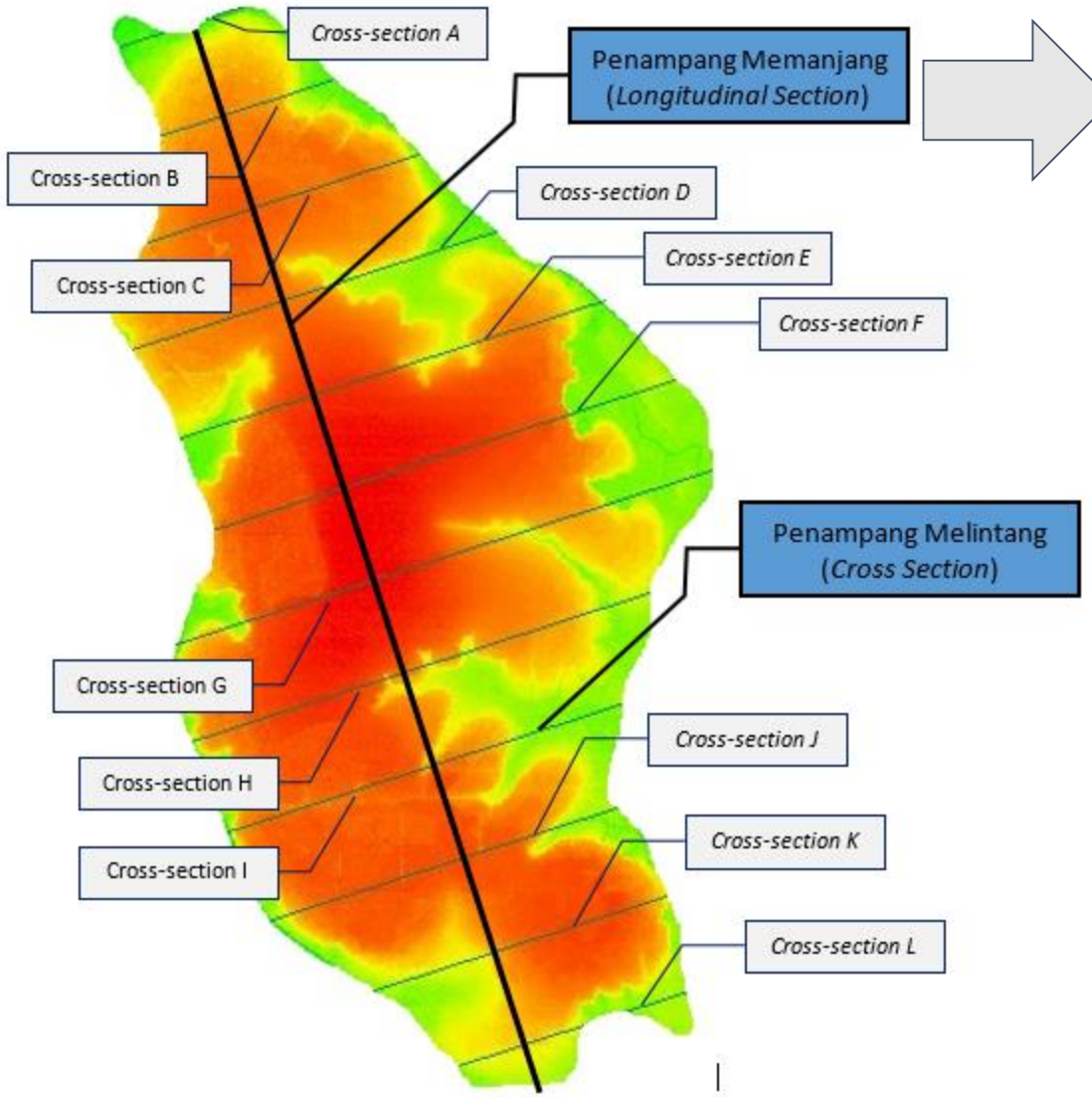


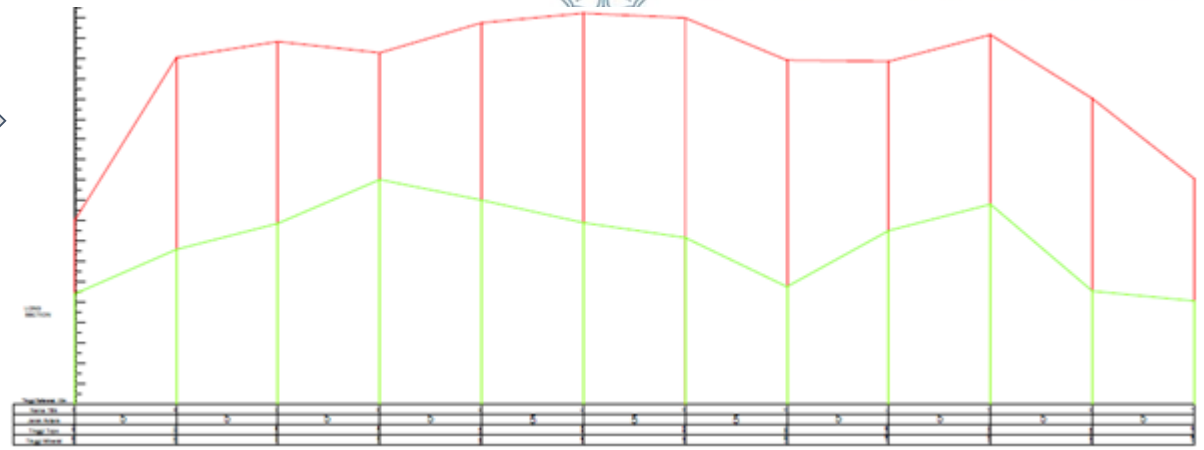
Pasang Surut

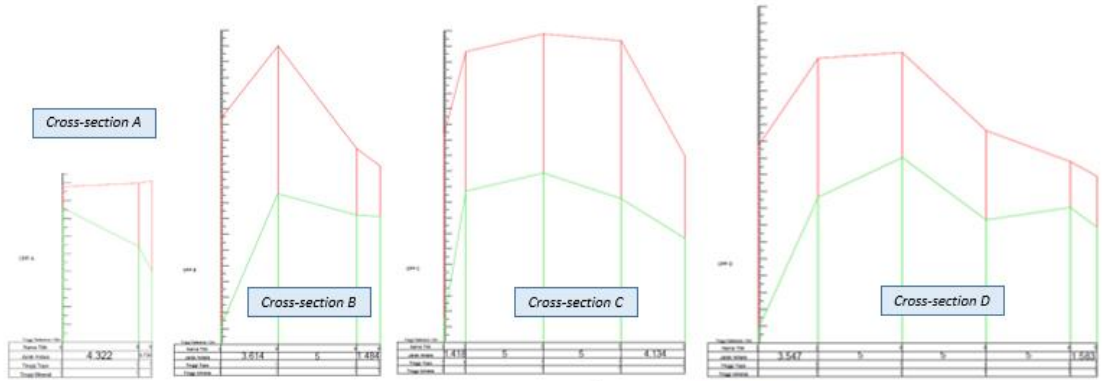


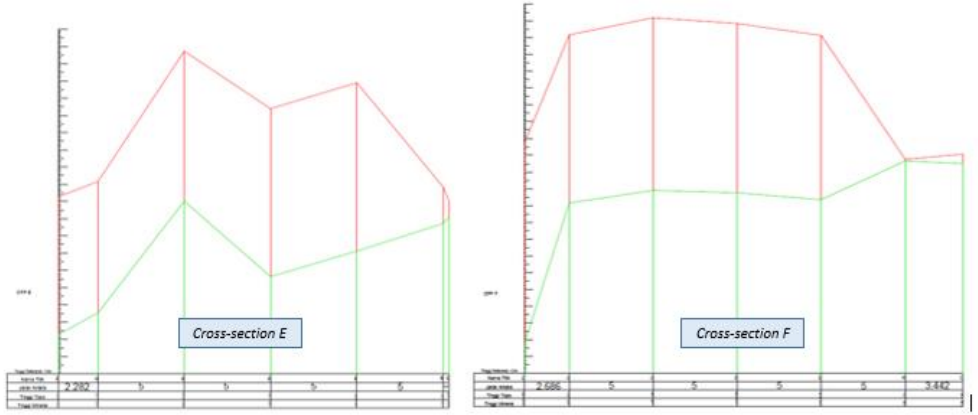


Analisis Kubah Gambut



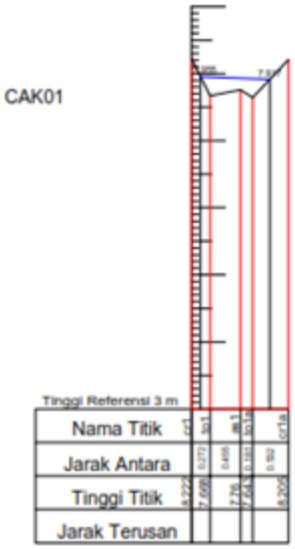




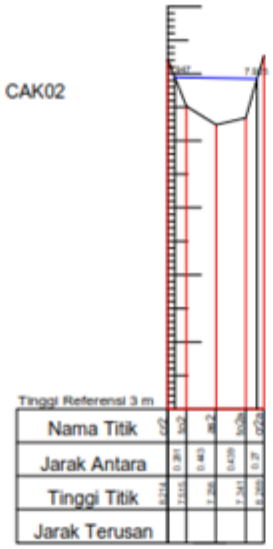


Analisis

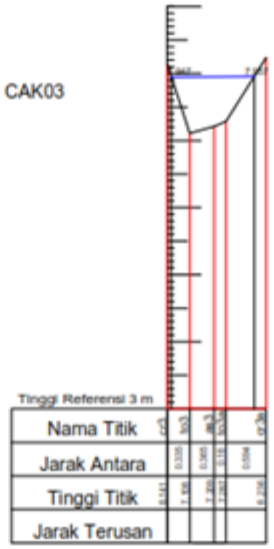
CAK01



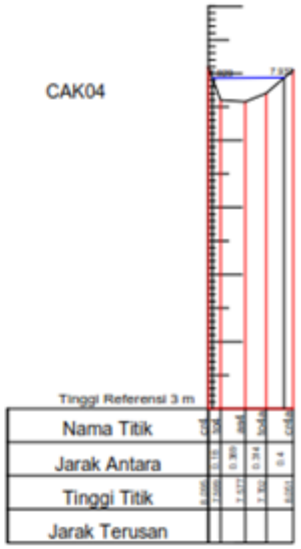
CAK02



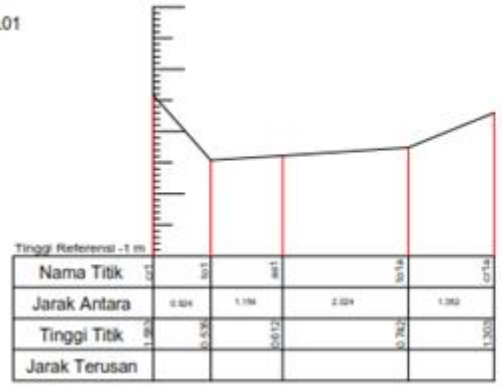
CAK03



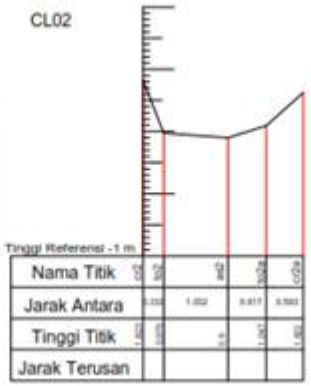
CAK04



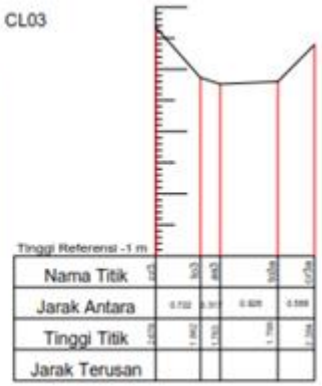
CL01



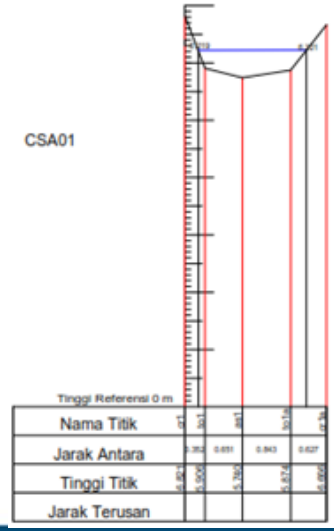
CL02



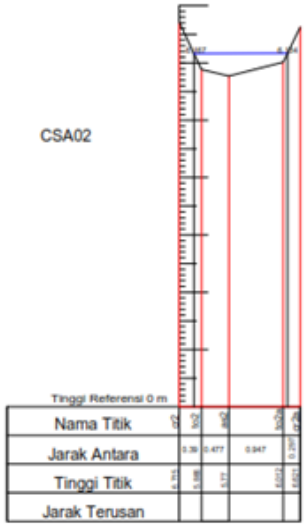
CL03



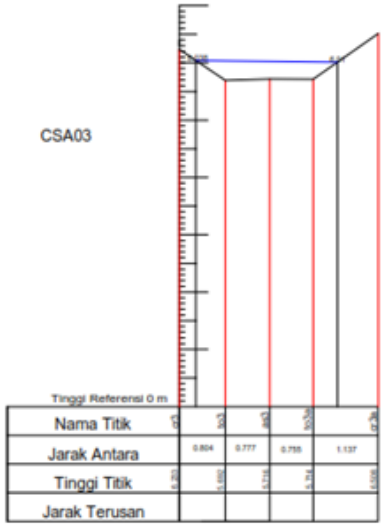
CSA01



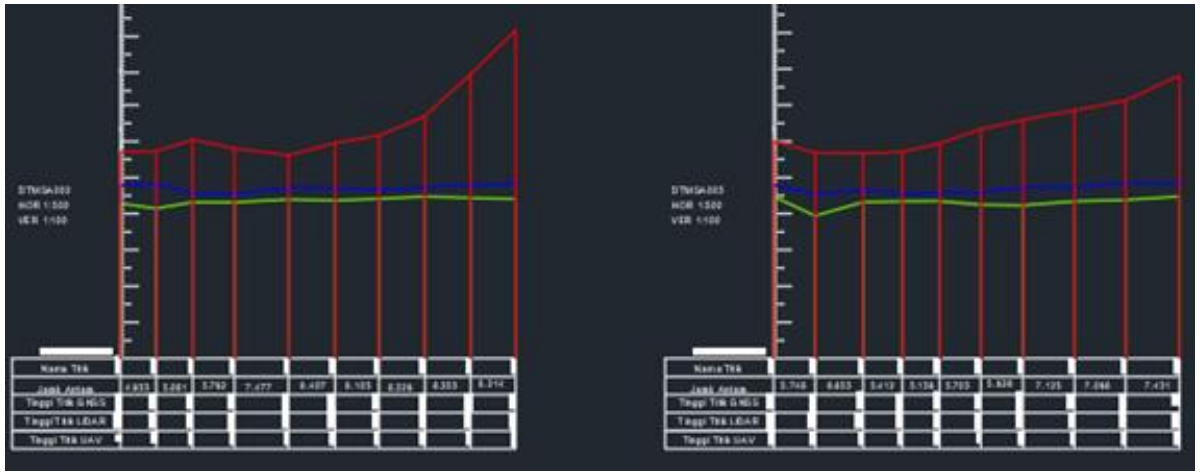
CSA02



CSA03



Analisis Akurasi



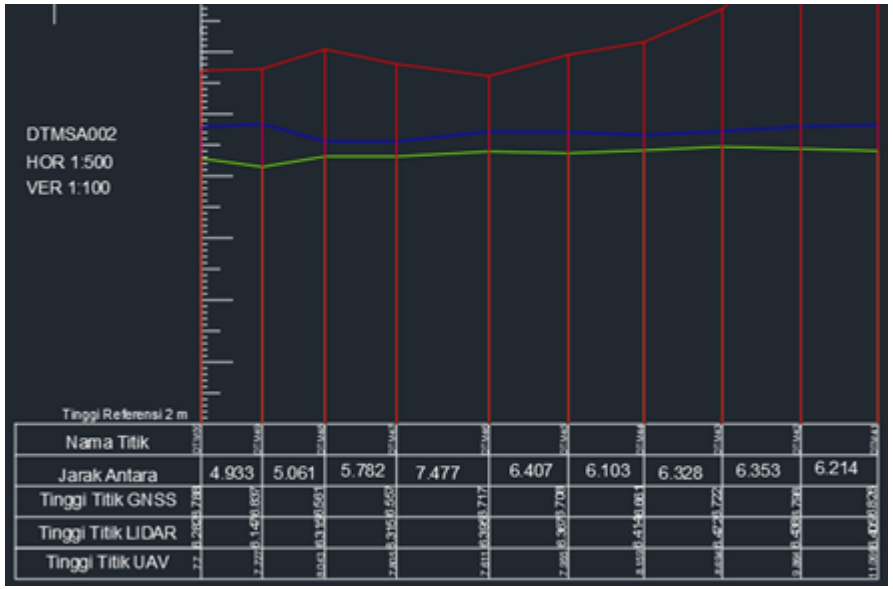
Quality Level (QL)	$RMSE_z$ (nonvegetated) (cm)	NVA at 95-percent confidence level (cm)	VVA at 95th percentile (cm)
QL0	≤5.0	≤9.8	≤14.7
QL1	≤10.0	≤19.6	≤29.4
QL2	≤10.0	≤19.6	≤29.4
QL3	≤20.0	≤39.2	≤58.8

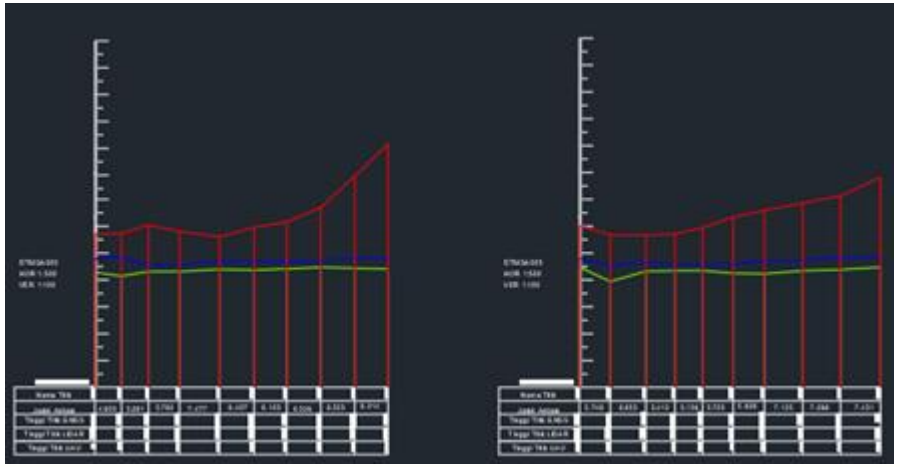
no	area	rata-rata (m)	RMSE (m)	LE-90	keterangan
1	area terbuka	0.042	0.206	0.339	non-vegetated
2	area pemukiman	0.036	0.191	0.315	non-vegetated
3	area semak	0.131	0.361	0.596	vegetated
4	area vegetasi rendah	0.259	0.509	0.840	vegetated
5	area hutan	0.294	0.543	0.895	vegetated
6	area rawa	0.164	0.405	0.668	vegetated
7	**keseluruhan area**	**0.120**	**0.347**	**0.572**	**over-all**

Analisis Akurasi UAV

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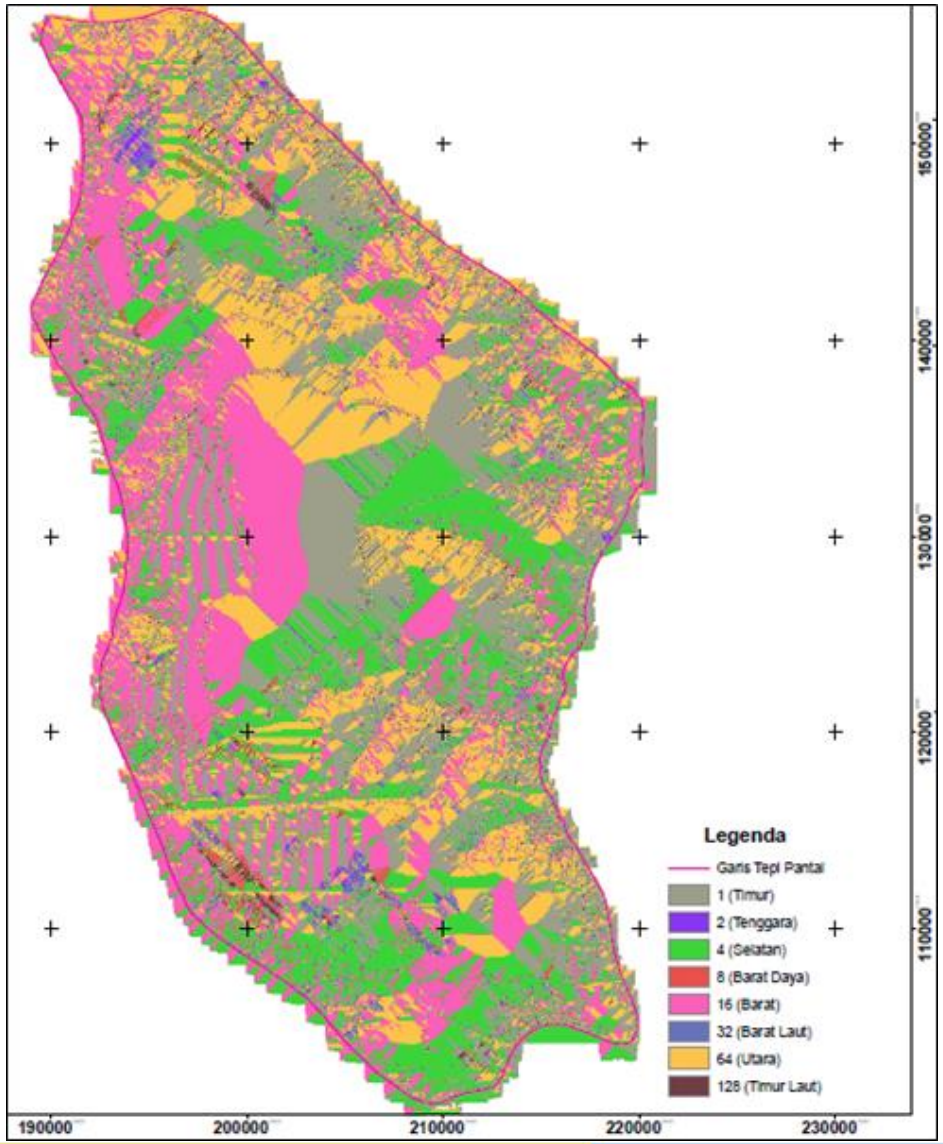
ICP	Ortho LIDAR		Ortho UAV			Difference	
	x	y	x	y		X	Y
CP1	219743.071	132380.774	219742.8	132380.842	JLN	0.271	0.068
CP2	220104.584	132627.894	220104.686	132628.555	BGN	0.102	0.661
CP3	219428.157	132771.117	219427.757	132772.07	BGN	0.4	0.953
CP5	219244.756	133119.63	219244.685	133119.811	JLN	0.071	0.181
CP6	219383.539	134032.402	219383.454	134032.336	JLN	0.085	0.066
CP7	219171.209	133469.329	219171.307	133469.543	JLN	0.098	0.214
CP8	218887.943	133233.314	218887.86	133233.213	LPGN	0.083	0.101
CP9	217961.205	133433.366	217960.874	133433.169	BGN	0.331	0.197
CP10	217338.317	133753.405	217338.295	133753.308	JLN	0.022	0.097
CP11	217674.403	134057.745	217674.42	134057.724	JLN	0.017	0.021
CP12	217104.737	134081.851	217104.798	134081.916	BGN	0.061	0.065
CP13	216759.433	134466.239	216759.363	134465.932	BGN	0.07	0.307
CP14	216573.524	134512.999	216573.553	134513.063	JLN	0.029	0.064
CP15	216310.929	134500.506	216310.91	134500.033	BGN	0.019	0.473
CP16	216042.476	134845.832	216042.571	134845.348	JLN	0.095	0.484
CP17	215693.893	134704.711	215693.927	134705.127	BGN	0.034	0.416
CP18	215005.858	134772.54	215005.872	134772.997	BGN	0.014	0.457
CP20	217725.083	133493.857	217725.513	133494.164	JLN	0.43	0.307
CP21	214558.039	134010.889	214557.513	134010.298	BGN	0.526	0.591
CP22	214042.729	133638.942	214042.196	133639.598	BGN	0.533	0.656
CP23	213625.618	133399.689	213624.845	133399.455	BGN	0.773	0.234
CP24	213313.12	133087.381	213313.275	133086.874	BGN	0.155	0.507
CP25	213774.879	132510.446	213773.559	132511.164	BGN	1.32	0.718
CP27	213111.954	133012.28	213111.988	133011.735	BGN	0.034	0.545
CP28	213152.64	133106.341	213153.858	133105.413	BGN	1.218	0.928
CP29	213014.072	133017.394	213014.424	133017.137	BGN	0.352	0.257
CP30	212830.393	132893.9	212830.721	132894.242	BGN	0.328	0.342
CP32	212067.669	134585.48	212068.516	134584.36	JLN	0.847	1.12
CP33	213931.359	133489.529	213931.135	133489.413	BGN	0.224	0.116
CP34	219748.907	132802.773	219749.119	132803.386	BGN	0.212	0.613
CP35	214265.371	133805.438	214265.121	133805.599	BGN	0.25	0.161
CP36	214414.443	133836.969	214414.517	133836.934	BGN	0.074	0.035
CP37	214725.978	134259.923	214726.522	134260.541	BGN	0.544	0.618
CP38	214971.045	134719.345	214971.395	134720.197	BGN	0.35	0.852
CP39	216133.142	134695.757	216132.912	134696.428	BGN	0.23	0.671
CP40	219468.29	133660.549	219467.535	133660.578	BGN	0.755	0.029
					AVERAGE	0.304361	0.392361





No	Kluster	Rerata Perbedaan Elevasi (m)
1	Rawa	1,685
2	Hutan Alami	8,325
3	Rumput	2,621
4	Area Terbuka	0,975
5	Pemukiman	3,068
6	Vegetasi Rendah	3,504

Analisis Arah Aliran & Akumulasi



78	72	69	71	58	49
74	67	56	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

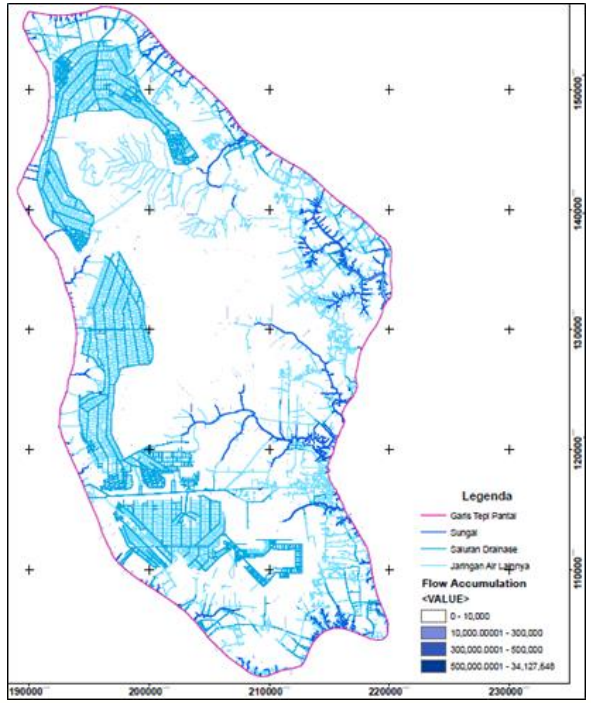
Elevation surface

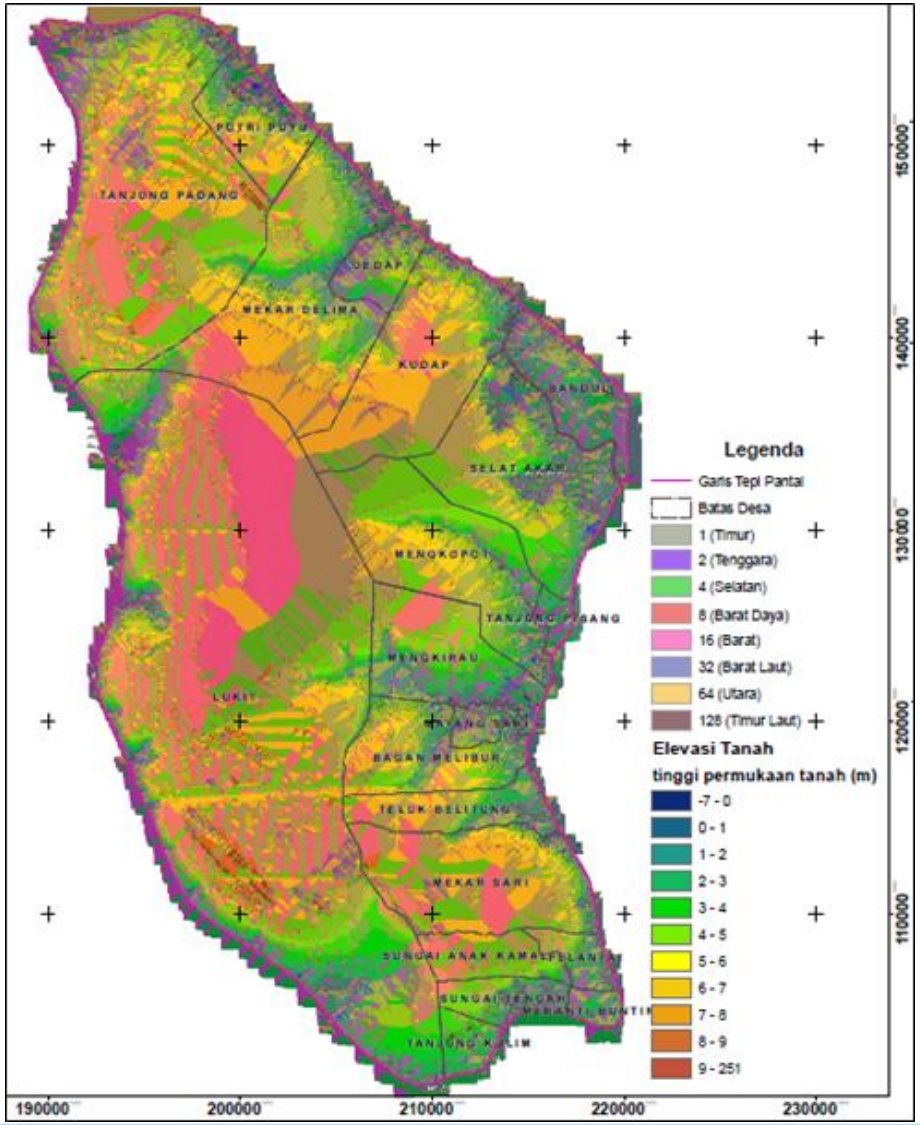
2	2	2	4	4	8
2	2	2	4	4	8
1	1	2	4	8	4
128	128	1	2	4	8
2	2	1	4	4	4
1	1	1	1	4	16

Flow direction

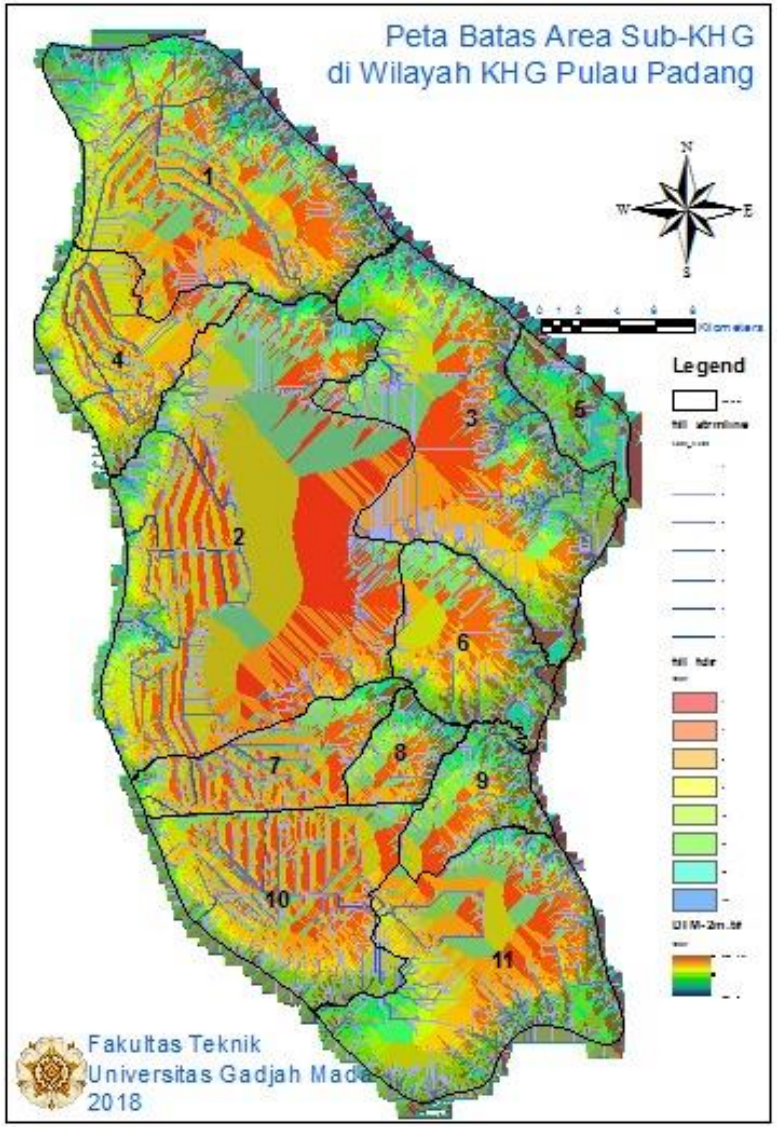
32	64	128
16		1
8	4	2

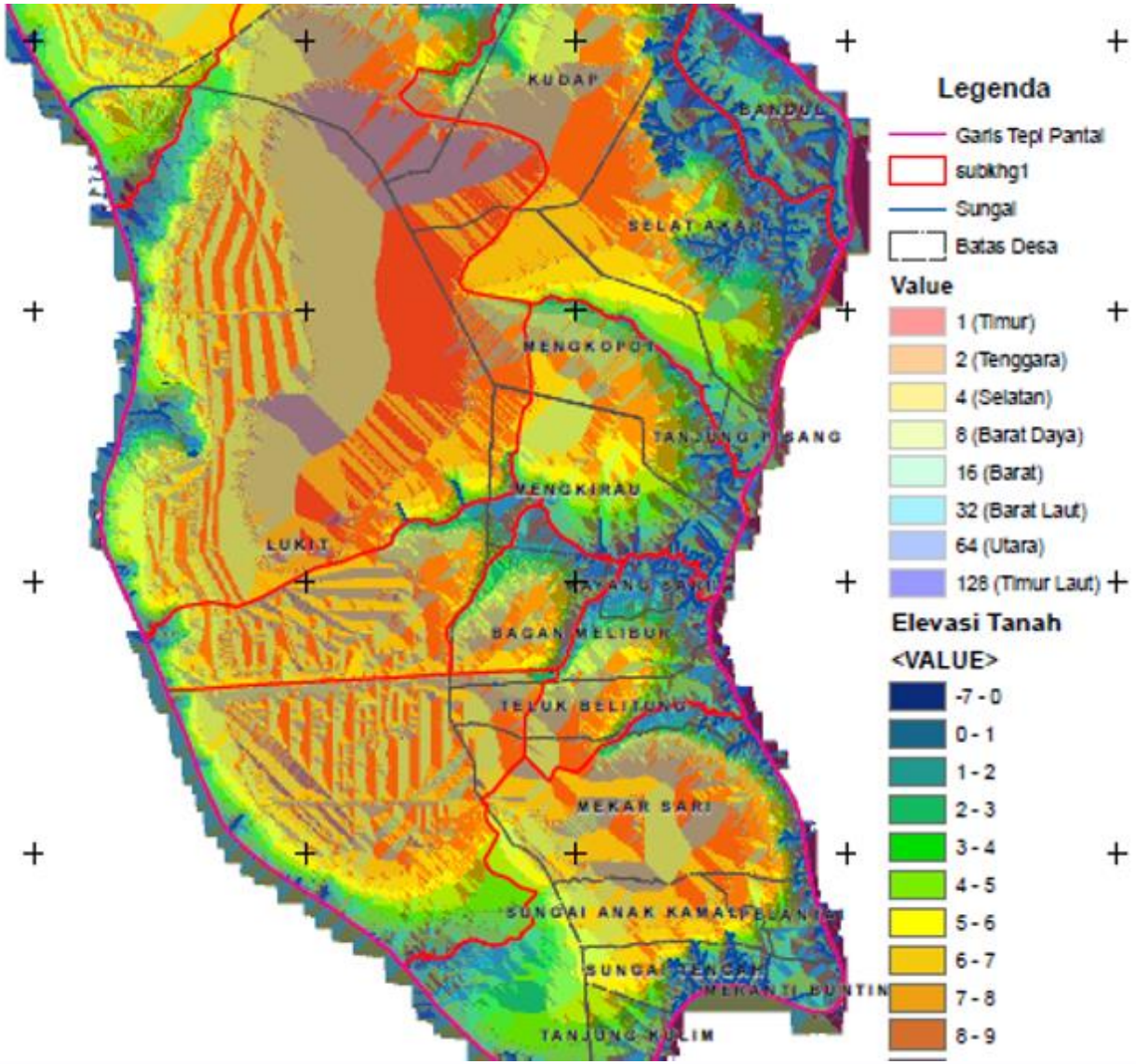
Direction coding





Analisis Sub KHG

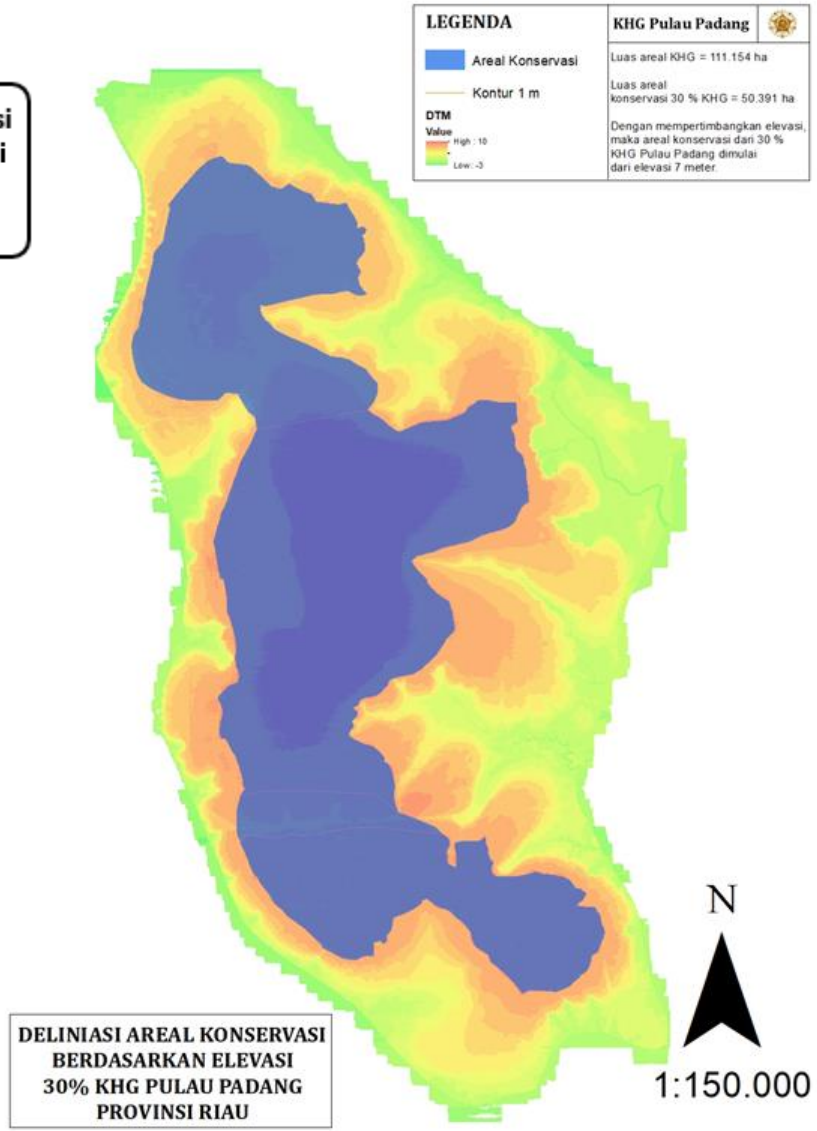




Analisis Penetapan Area Konservasi

Berdasarkan Elevasi

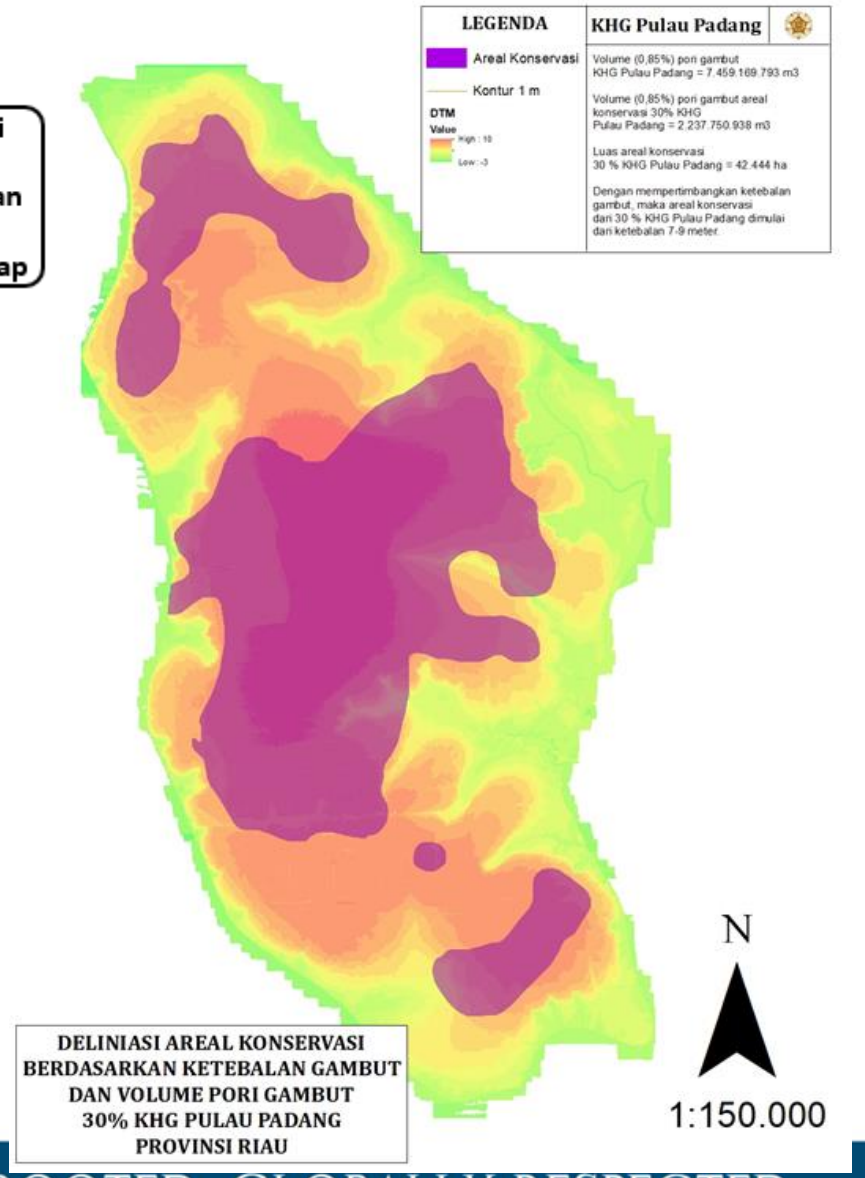
DTM LIDAR → Ekstraksi kontur 1m → Digitasi kontur per 1m menjadi luasan → Hitung luas KHG dan SubKHG → **Presentase fungsi lindung 30% dari luasan kontur yang overlap**



Analisis Penetapan Area Konservasi

Berdasarkan Ketebalan Gambut

DTM Ketebalan Gambut → Digitasi area ketebalan per 1m → Hitung luas dan volume area ketebalan → Intersect area ketebalan dengan deliniasi KHG dan SubKHG → **Presentase fungsi lindung 30% dari hitungan luasan dan volume tebal gambut yang overlap**

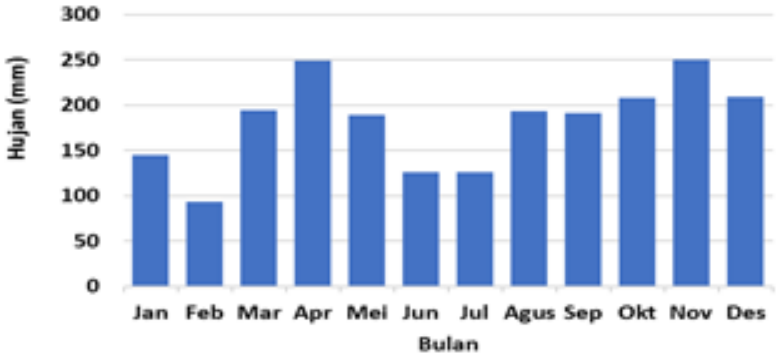


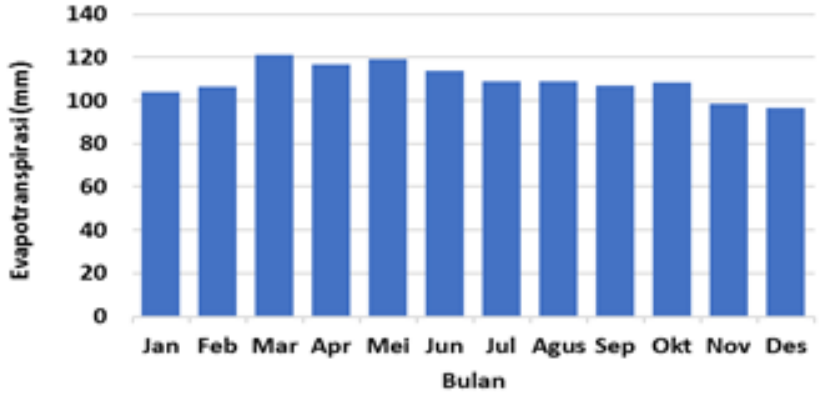
Hujan & Evapotransporasi & Neraca Air Global per SubKHG

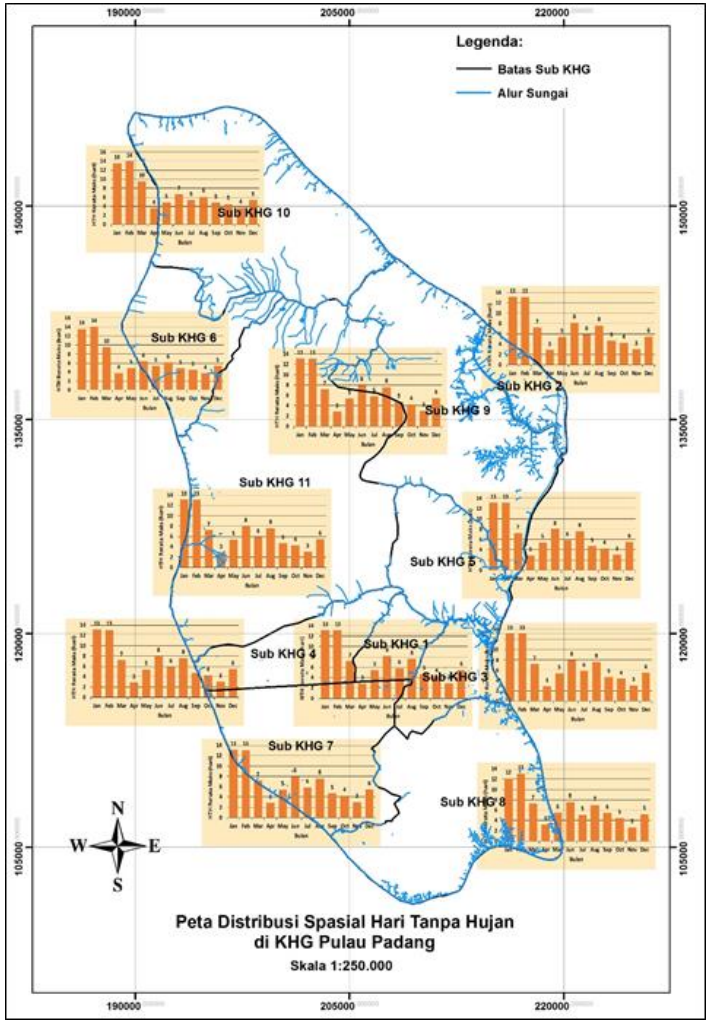
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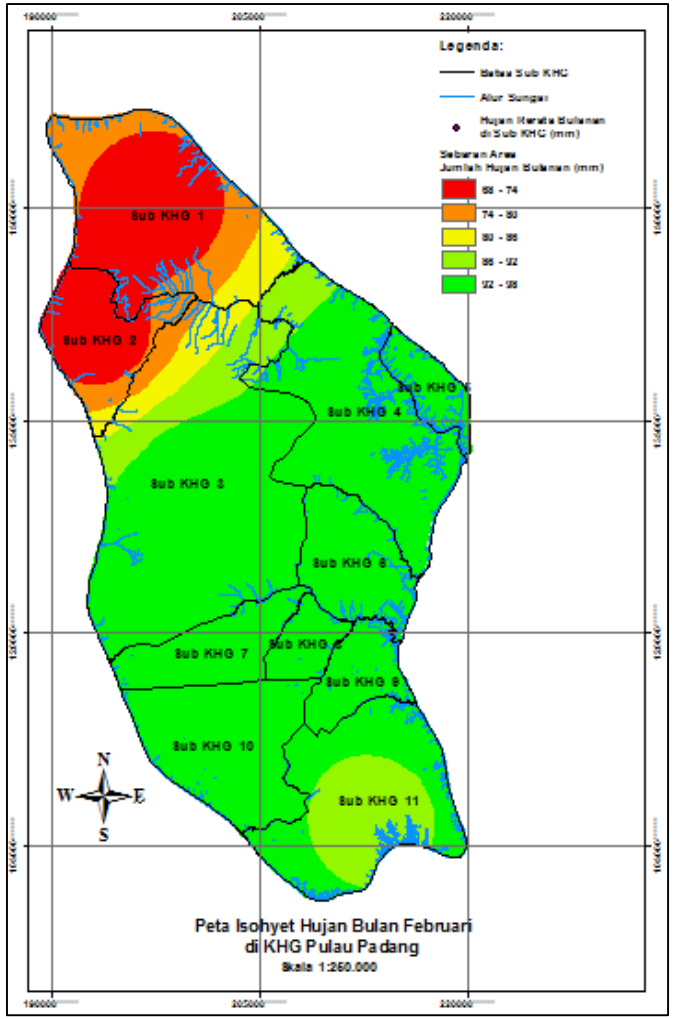
Sumber data: data satelit dan data sekunder perusahaan

Parameter	Nilai Parameter Neraca Air Global (mm)											
	Jan	Feb	Mar	Apr	Mei	Jun	Jul	Agus	Sep	Okt	Nov	Des
Hujan (P)	199	117	203	263	199	154	171	199	219	246	291	257
Evapotranspirasi (ET)	107	109	124	119	120	114	110	110	108	110	101	99
Runoff sungai (RO)	30	17	30	40	30	23	26	30	33	37	44	38
Status NA	62	-10	49	105	49	16	35	59	78	99	147	119



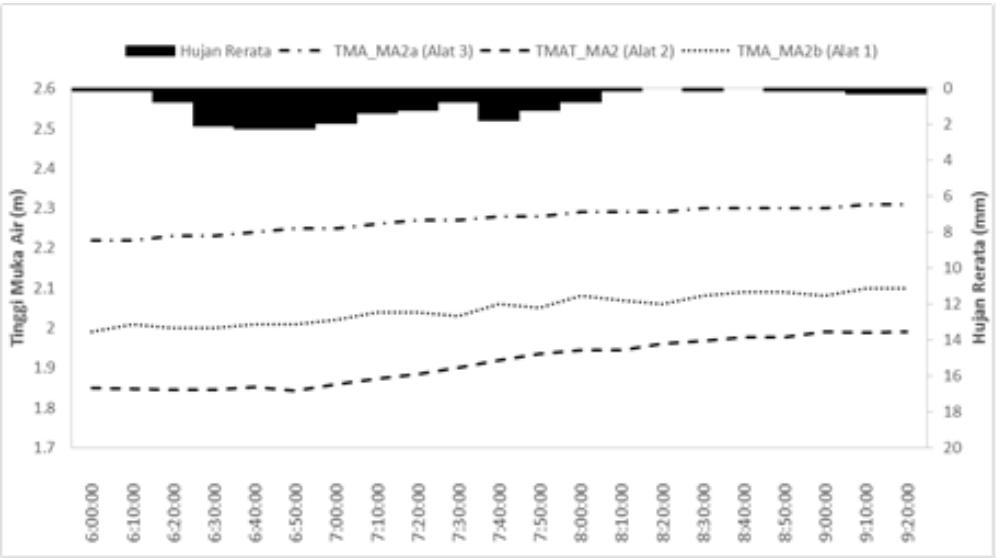


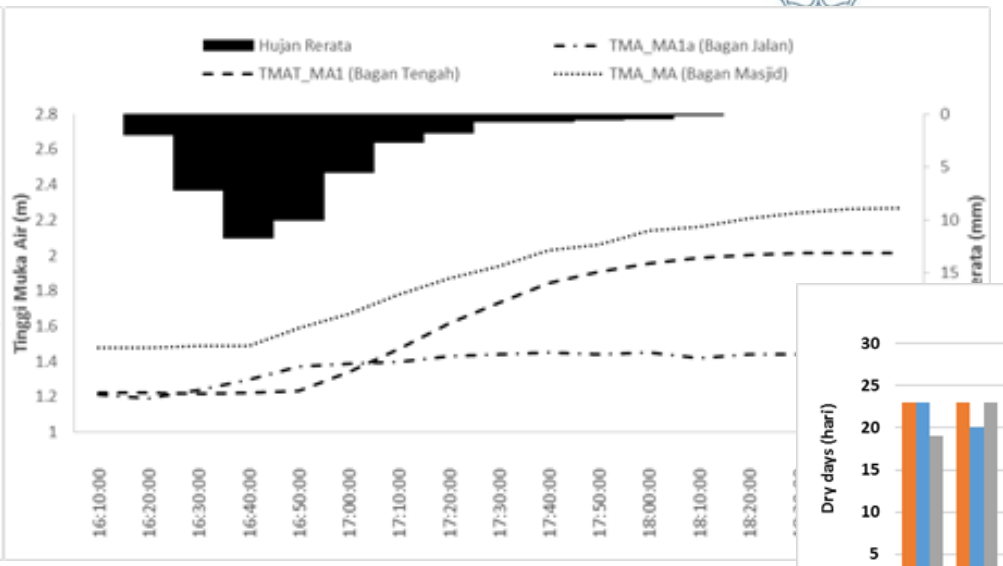


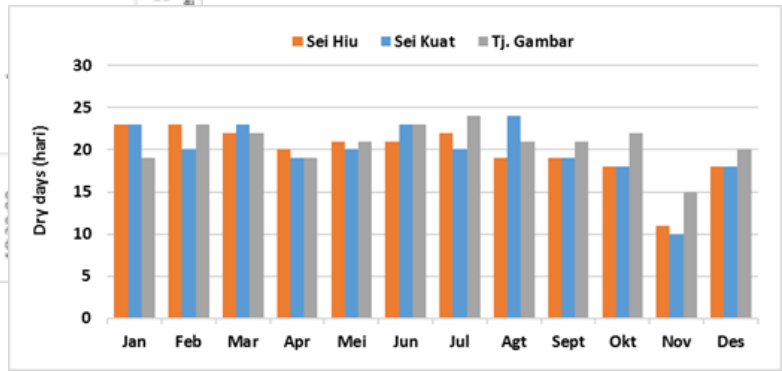


Analisis Curah Hujan & Dry Days

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Lokasi Sei Hiu

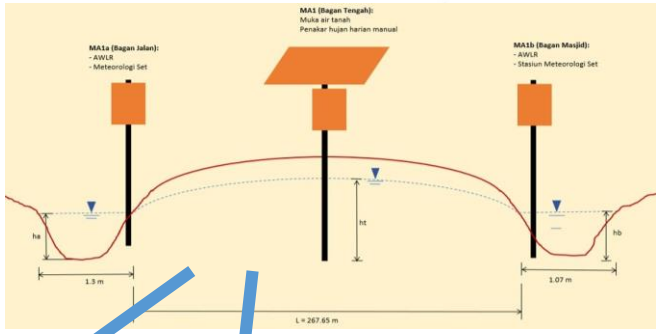
Tahun	Parameter	Jan	Feb	Mar	Apr	Mei	Jun	Jul	Agt	Sept	Okt	Nov	Des	Jumlah
2014	Total (mm)	77	-	95	241	136	102	105	213	126	212	290	439	2035
	Max (mm)	34	-	49	58	27	39	51	34	25	52	50	68	
	Rain days (hr)	5	-	4	12	13	6	6	17	9	11	19	17	
	Dry days (hr)	26	-	27	18	18	24	25	14	21	20	11	14	
2015	Total (mm)	41	87	76	137	55	219	18	179	69	108	268	255	1512
	Max (mm)	27	83	29	62	19	97	7	107	24	30	97	63	
	Rain days (hr)	4	2	6	10	4	9	4	5	6	9	18	13	
	Dry days (hr)	27	26	25	20	27	21	27	26	24	22	12	18	
2016	Total (mm)	42	44	84	29	201	147	196	243	145	141	266	126	1664
	Max (mm)	9	24	61	25	58	35	47	63	33	24	63	24	
	Rain days (hr)	6	3	5	3	12	10	16	9	11	17	21	9	
	Dry days (hr)	25	26	26	27	19	20	15	22	19	14	9	22	
2017	Total (mm)	319	83	278	284	151	226	48	213	292	191	-	-	2083
	Max (mm)	125	19	103	102	59	55	29	45	51	40	-	-	
	Rain days (hr)	16	11	22	15	9	10	4	18	18	15	-	-	
	Dry days (hr)	15	17	9	16	22	20	27	13	12	16	-	-	
Hujan bulanan (mm)		119	71	133	173	136	173	92	212	158	163	275	273	1824

Lokasi Sei Kuat

Tahun	Parameter	Jan	Feb	Mar	Apr	Mei	Jun	Jul	Agt	Sept	Okt	Nov	Des	Jumlah
2014	Total (mm)	72	-	32	249	268	97	142	169	90	306	271	174	1870
	Max (mm)	27	-	12	107	66	62	37	102	29	72	64	40	
	Rain days (hr)	4	-	4	9	12	4	15	8	10	13	22	12	
	Dry days (hr)	27	-	27	21	19	26	16	23	20	18	8	19	
2015	Total (mm)	27	13	33	174	124	155	86	108	36	157	211	175	1299
	Max (mm)	14	9	15	49	49	60	20	34	14	41	60	45	
	Rain days (hr)	3	2	4	13	7	5	7	7	5	9	18	14	
	Dry days (hr)	28	26	27	17	24	25	24	24	25	22	12	17	
2016	Total (mm)	192	45	21	45	231	88	155	130	238	134	416	137	1831
	Max (mm)	53	25	10	22	44	30	29	56	64	53	76	41	
	Rain days (hr)	11	5	3	5	10	10	14	5	11	15	21	13	
	Dry days (hr)	20	24	28	25	21	20	17	26	19	16	9	14	
2017	Total (mm)	217	183	194	235	312	159	93	147	211	236	-	-	1986
	Max (mm)	76	53	60	92	72	41	37	81	35	91	-	-	
	Rain days (hr)	14	17	20	15	14	10	6	8	18	13	-	-	
	Dry days (hr)	17	11	11	15	17	20	25	23	12	18	-	-	
Hujan bulanan (mm)		127	80	70	176	234	125	119	139	144	208	300	162	1747

Sumber: data sekunder perusahaan

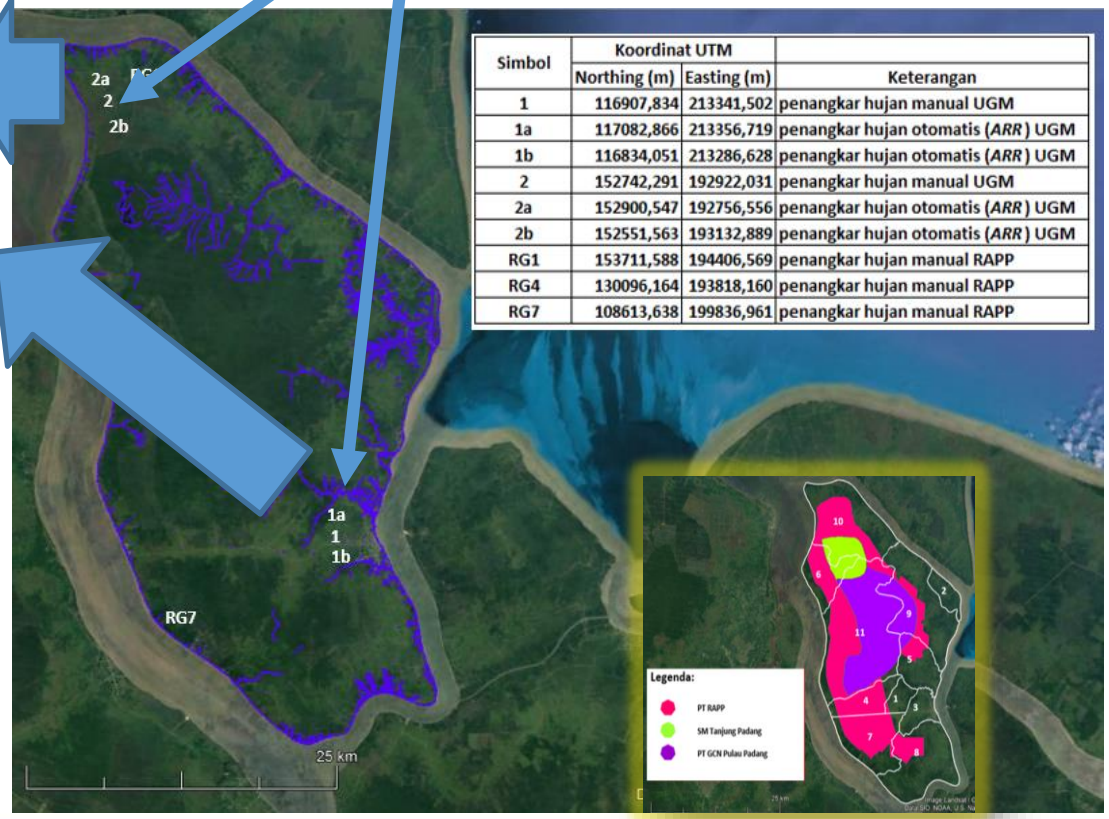
Analisis Neraca Air Rinci



Lokasi	Tanggal	Total hujan (P)			Luas Lahan	Vol P_{rerata}	Vol. RO	Vol. ET	dS	RSE
		Alat 3	Alat 2	Alat 1						
		mm	mm	mm	m^2	m^3	m^3	m^3	m^4	%
Desa Sei Hiu	10-Nov	2.7	5.8	12.6	1375557.9	9686.13	733.75	4547.92831	4404.45	45.5
	17-Nov	3.9	49.0	34.5	1375557.9	40086.23	4630.88	4547.92831	30907.42	77.1

Lokasi	Tanggal	Total hujan (P)			Luas Lahan	Vol P_{rerata}	Vol. RO	Vol. ET	dS	RSE
		B. Jalan	B. Tengah	B. Masjid						
		mm	mm	mm	m^2	m^3	m^3	m^3	m^4	%
Desa Bagan Melibur	6-Nov	39.6	48.3	6.9	130796	4133	684	425	3023	73.2
	7-8 Nov	49.2	45.3	89.6	130796	8027	895	425	6706	83.6
	19-Nov	21.0	26.4	46.2	130796	4081	210	425	3446	84.4
	3-Dec	31.5	36.6	52.4	130796	5255	354	416	4484	85.3
	8-Dec	12.3	10.5	-	130796	1491	187	416	888	59.6

Lokasi	Tanggal	Hujan (mm)	C_{low}	C_{peak}	C_{low} rerata	C_{peak} rerata	C_{desain}
Bagan Melibur	6 Nov. 2017	43.7	0.120	0.165	0.130	0.209	0.25
	7 Nov. 2017	42.9	0.160	0.311			
	3 Des. 2017	24.5	0.111	0.174			
	8 Des. 2017	11.1	0.129	0.187			
Sei Hiu Tanjung Padang	10 Nov. 2017	7.5	0.071	0.142	0.124	0.228	0.35
	17 Nov. 2017	19.1	0.177	0.314			

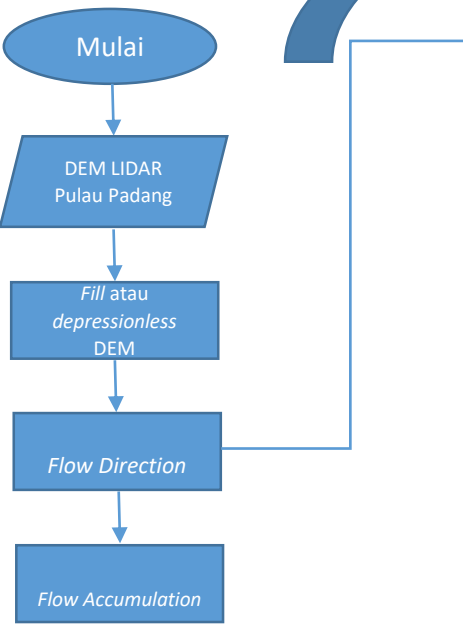


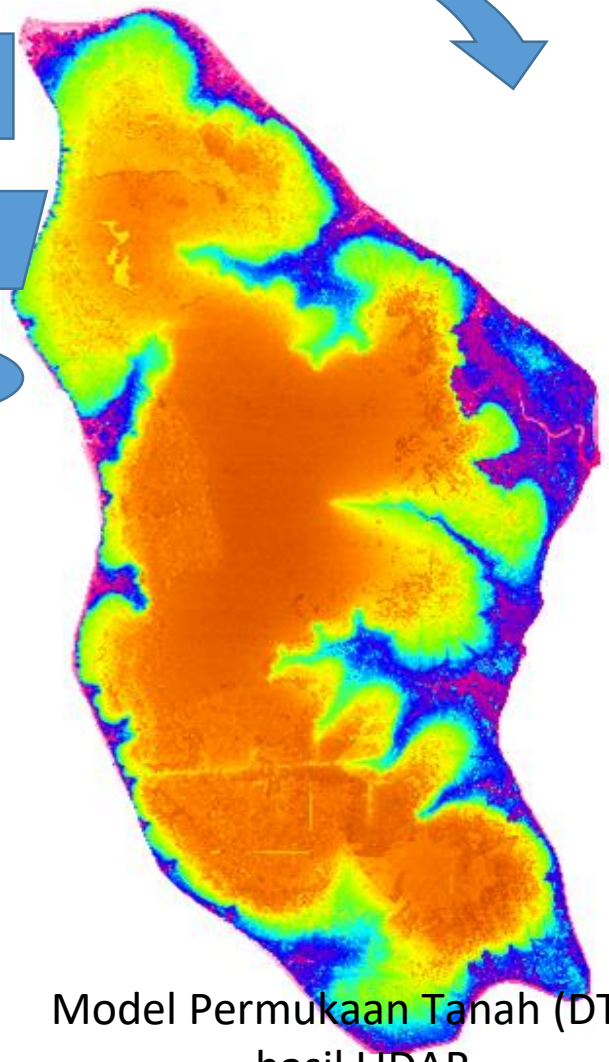
Simbol	Koordinat UTM		
	Northing (m)	Easting (m)	Keterangan
1	116907,834	213341,502	penangkar hujan manual UGM
1a	117082,866	213356,719	penangkar hujan otomatis (*ARR*) UGM
1b	116834,051	213286,628	penangkar hujan otomatis (*ARR*) UGM
2	152742,291	192922,031	penangkar hujan manual UGM
2a	152900,547	192756,556	penangkar hujan otomatis (*ARR*) UGM
2b	152551,563	193132,889	penangkar hujan otomatis (*ARR*) UGM
RG1	153711,588	194406,569	penangkar hujan manual RAPP
RG4	130096,164	193818,160	penangkar hujan manual RAPP
RG7	108613,638	199836,961	penangkar hujan manual RAPP

Pemanfaatan LIDAR untuk RG

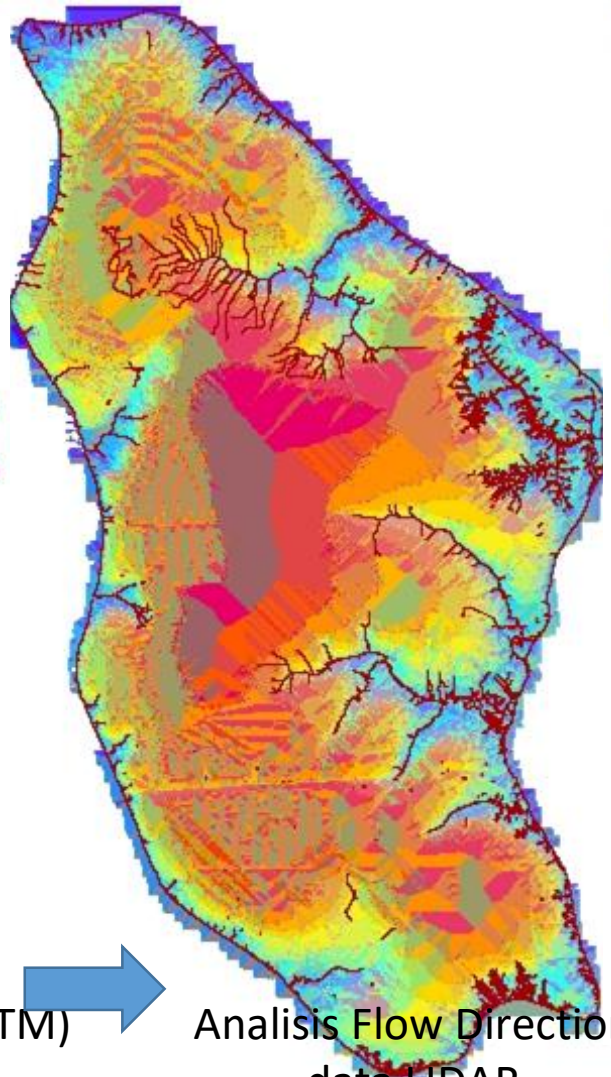
- Use of Lidar for Engineering Projects
  - Hydrological Modelling
    - Catchment Analysis (Stream Coverage (DNR, NHD) • DEM Reconditioning • Fill Sinks (depressionless DEM) • Flow Direction Grid • Flow Accumulation Grid • Define Stream Network • Drainage Line • Watershed Delineation)
    - Drainage Area Delineation
    - Flow Length
    - Time of Concentration
    - Floodplain Storage Volumes
    - Impoundment Site Volumes
    - Wetland Storage Volumes

Contoh Tindak Lanjut Hasil LIDAR: Menetapkan Batas KHG (Kasus P. Padang)

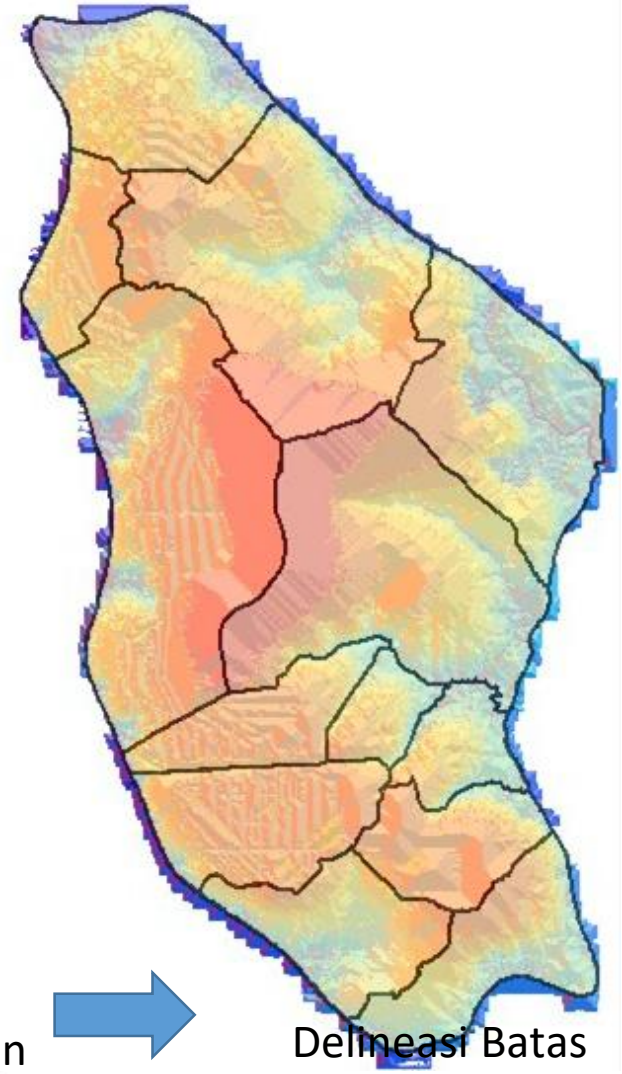




Model Permukaan Tanah (DTM) hasil LIDAR

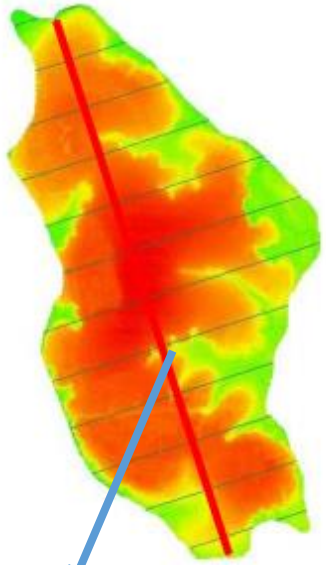


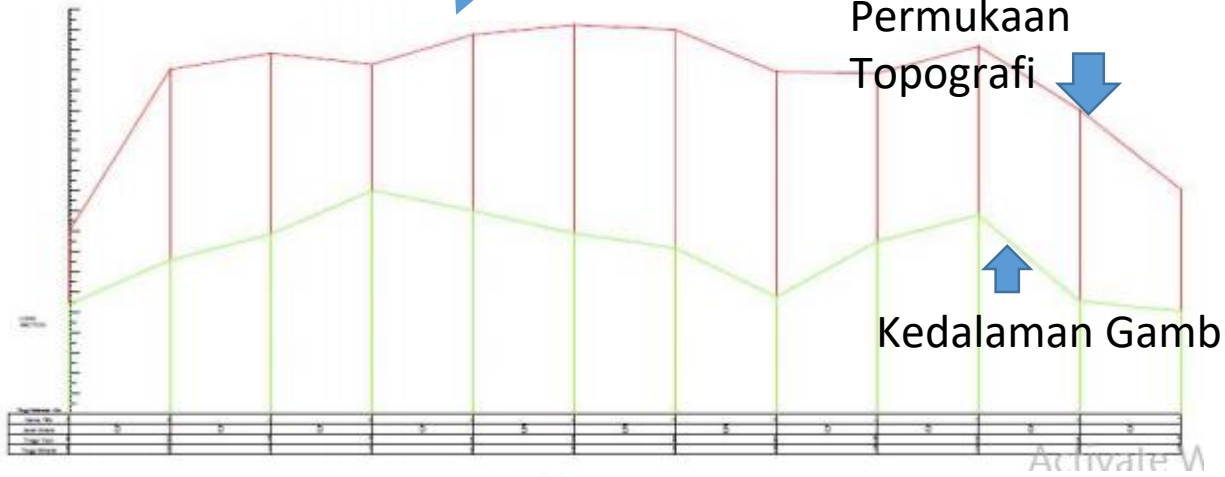
Analisis Flow Direction data LIDAR



Delineasi Batas KHG (Sub KHG)

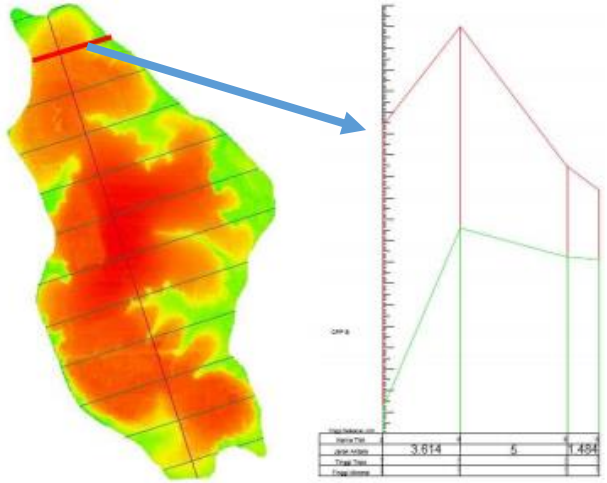
Hasil Survei Gambut vs. LIDAR



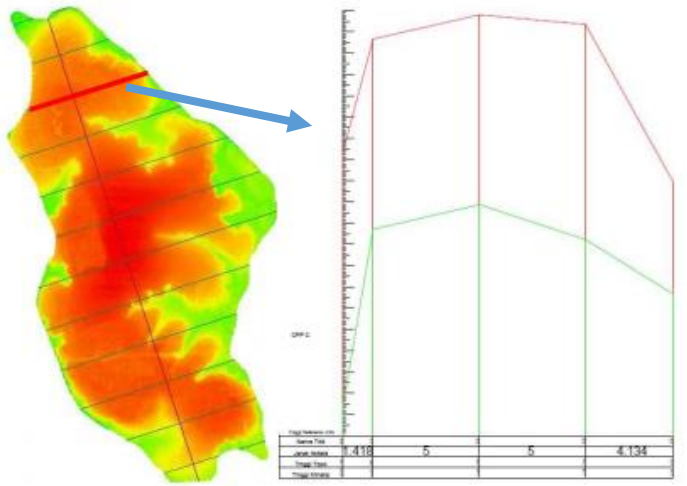


Permukaan Topografi

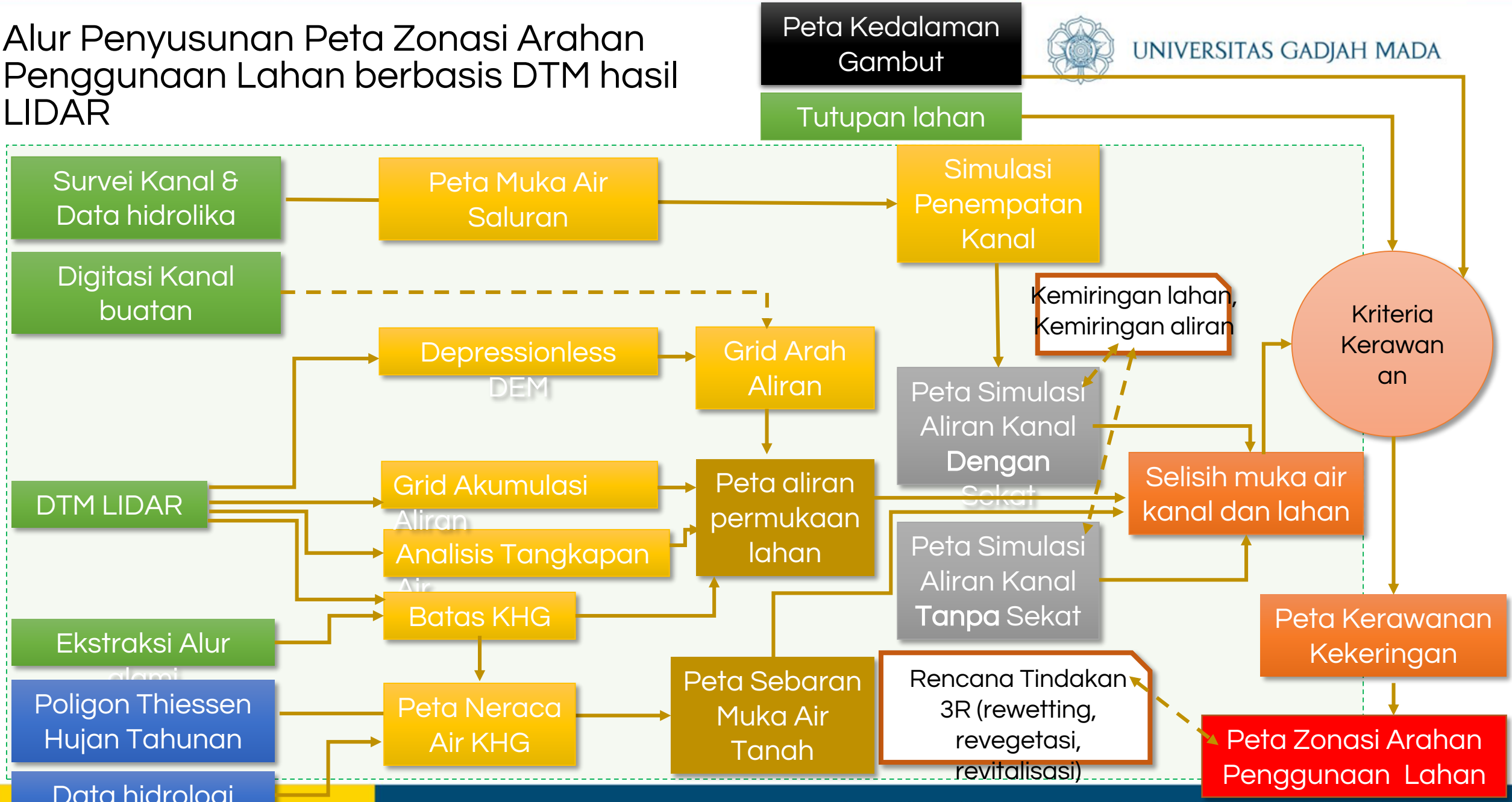
Kedalaman Gambut



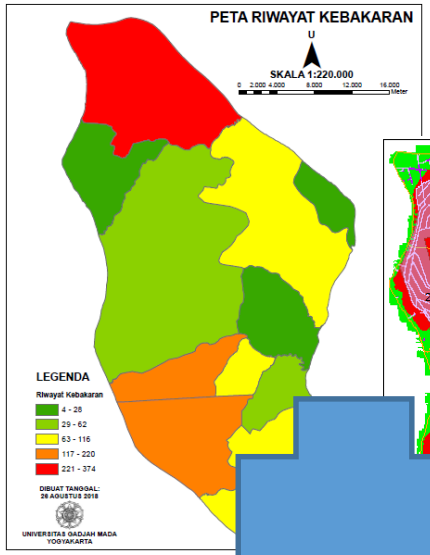
c. *Cross Section C*



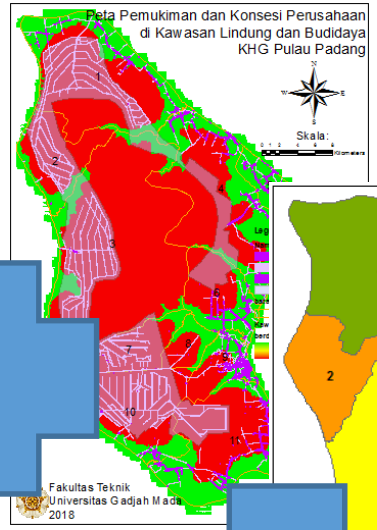
Alur Penyusunan Peta Zonasi Arahan Penggunaan Lahan berbasis DTM hasil LIDAR



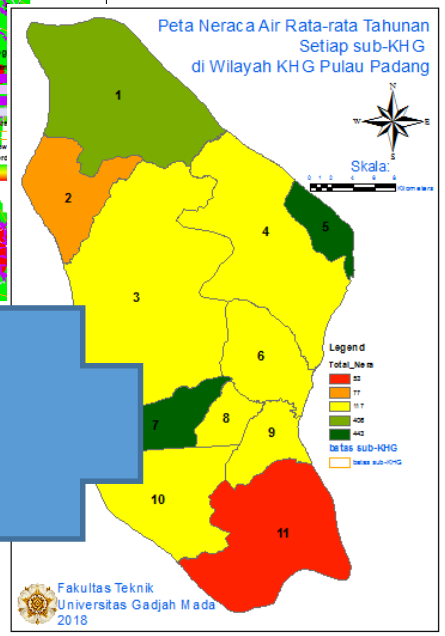
Peta Kerawanan Kering per Sub KHG



Riwayat Terbakar per SubKHG

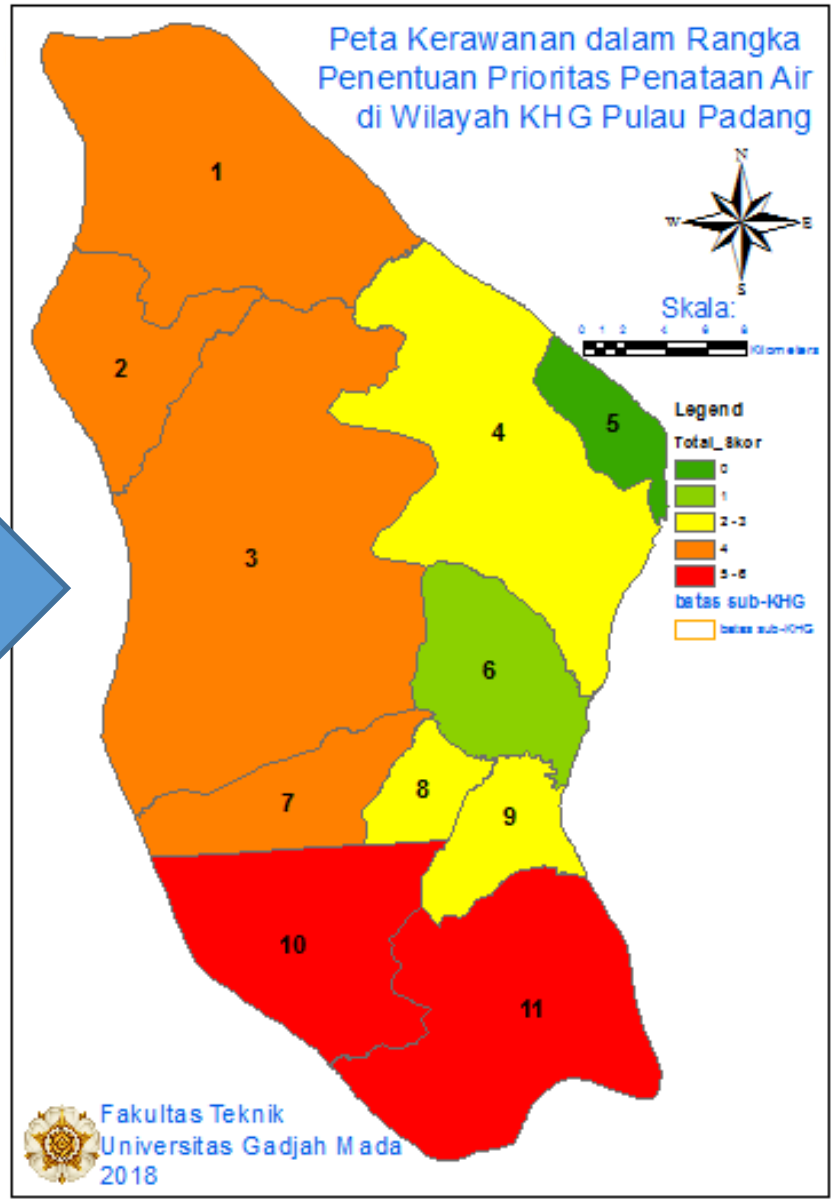


Budidaya & Gambut di atas 3 m

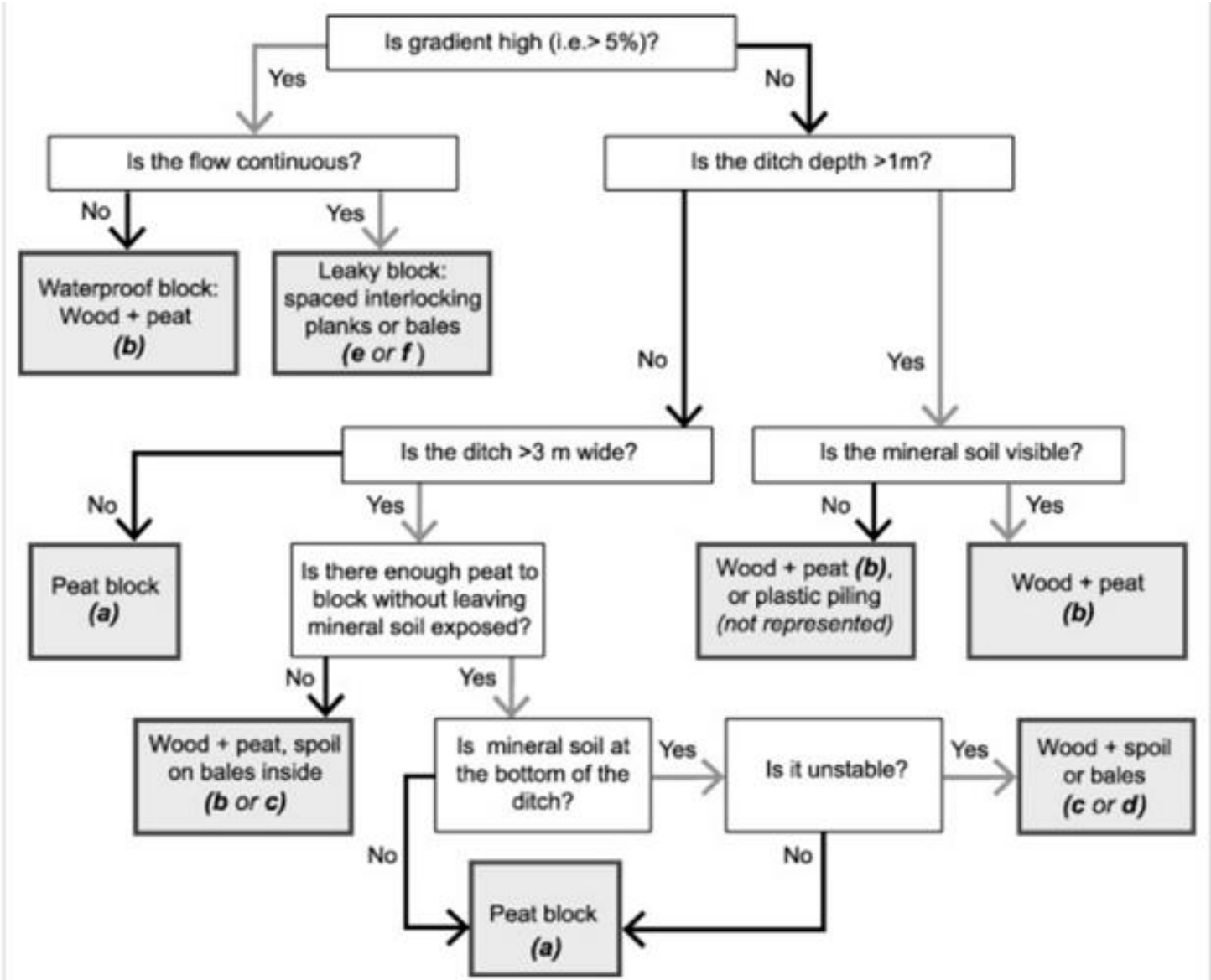


Neraca Air per Sub KHG

Berdasarkan Zona Peruntukan (lindung atau budidaya)		Berdasarkan Riwayat Kejadian Kebakaran		Berdasarkan Hitungan Neraca Air	
Zona budidaya	Score 0	Kejadian < 50	Score 0	Neraca air > 200	Score 0
Sebagian besar masuk zona budidaya	Score 1	50 < kejadian < 100	Score 1	100 < neraca air < 200	Score 1
Zona lindung	Score 2	200 > kejadian	Score 2	100 > neraca air	Score 2



Topography-Canalblocking-Peatdepth

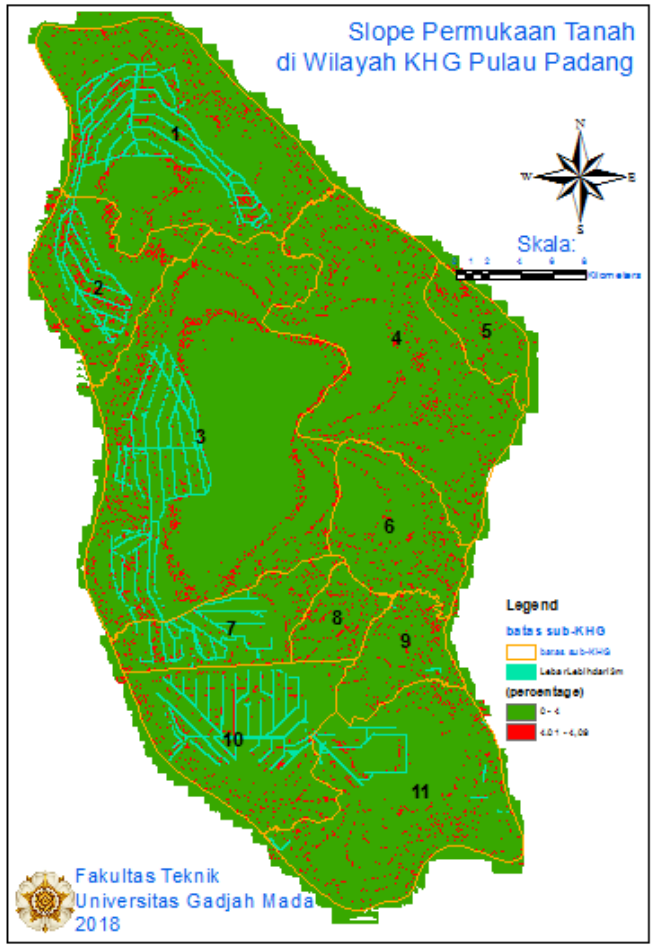


Komponen Informasi dari Lidar untuk Indikasi penyekatan

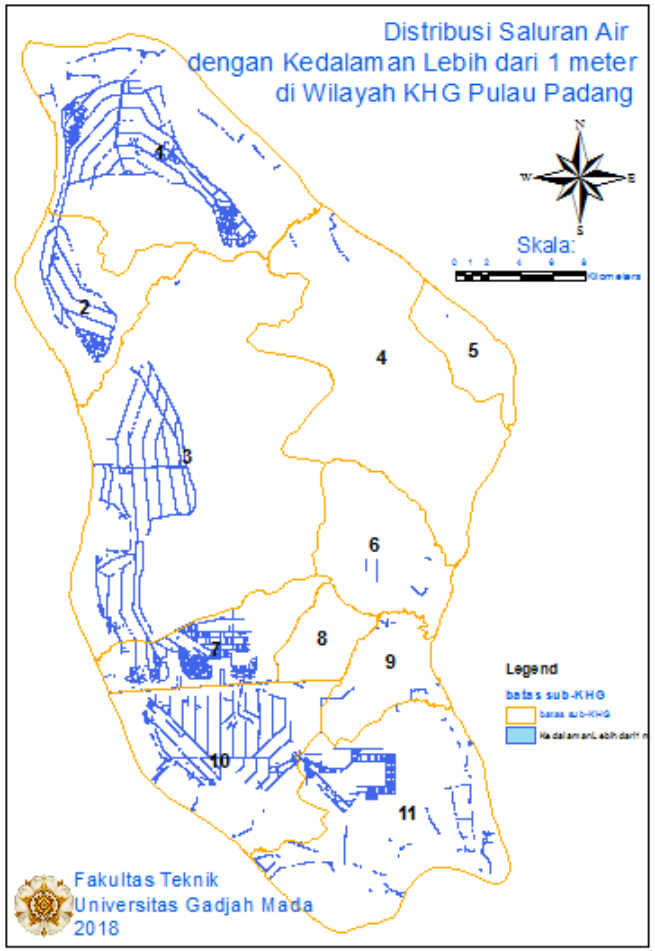
1. Gradien saluran > 5 %
2. Kedalaman saluran > 1 meter
3. Lebar saluran > 3 meter
4. Posisi saluran terhadap tanah mineral

E. Grand-Clement, K. Anderson, D. Smith, M. Angus, D.J. Luscombe, N. Gatis, L.S. Bray, R.E. Brazier, ***New approaches to the restoration of shallow marginal peatlands***, Journal of Environmental Management, Volume 161, 15 September 2015, Pages 417-430, ISSN 0301-4797,

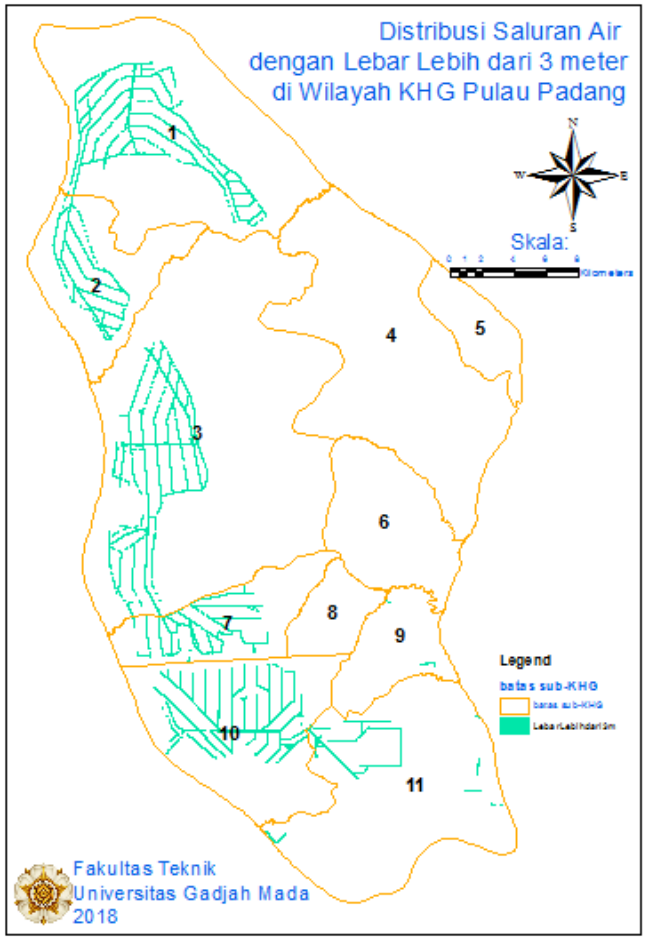
Topography-Canalblocking-Peatdepth



Slope → Gradien 5%



Kedalaman Saluran Air dari Lahan

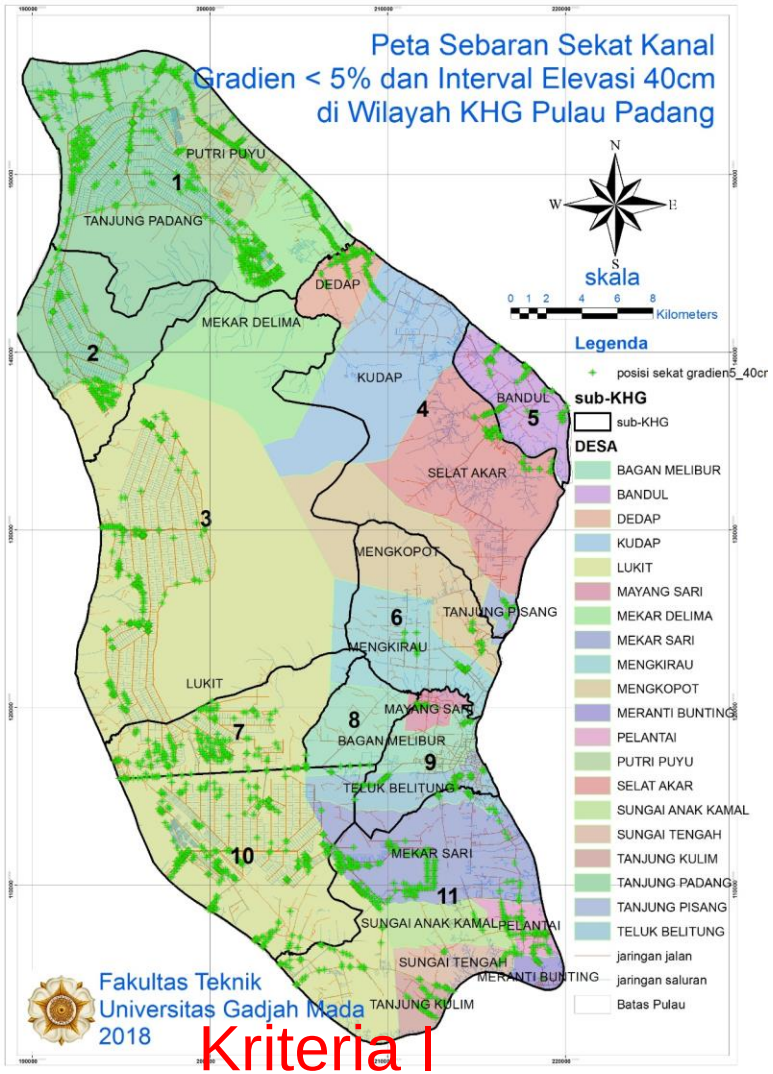


Lebar Saluran Air

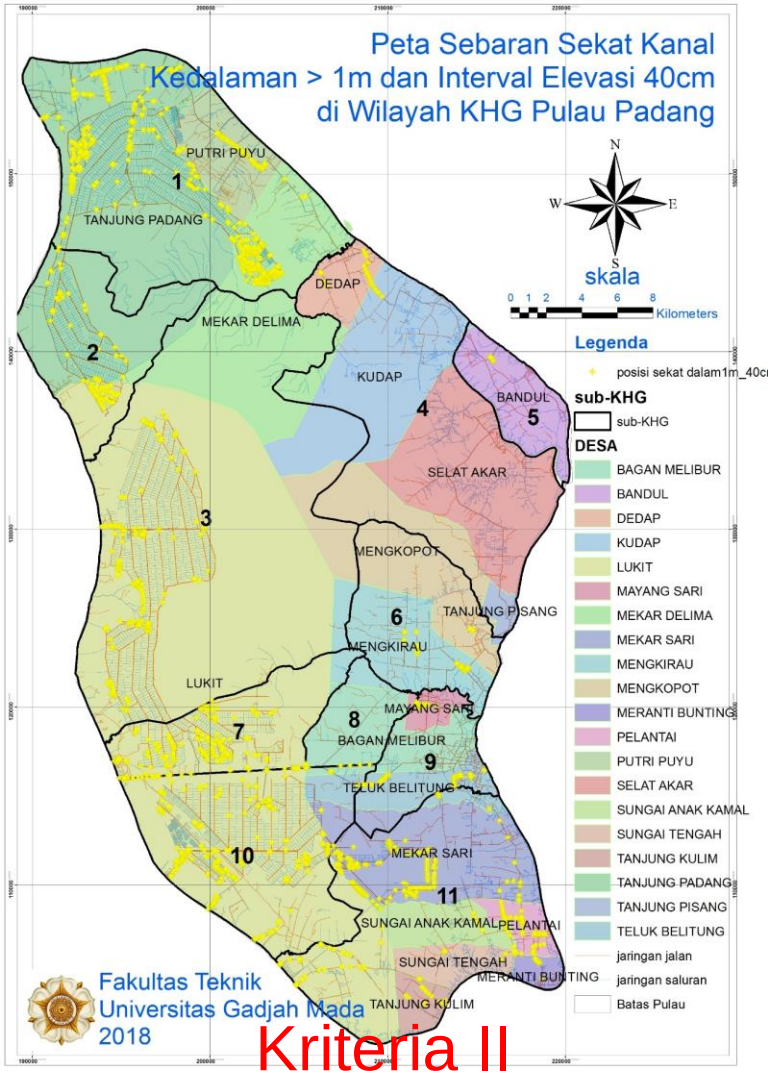
Komponen Informasi dari Lidar untuk Indikasi penyekatan

1. Gradien saluran > 5 %
2. Kedalaman saluran > 1 meter
3. Lebar saluran > 3 meter
4. Posisi saluran terhadap tanah mineral

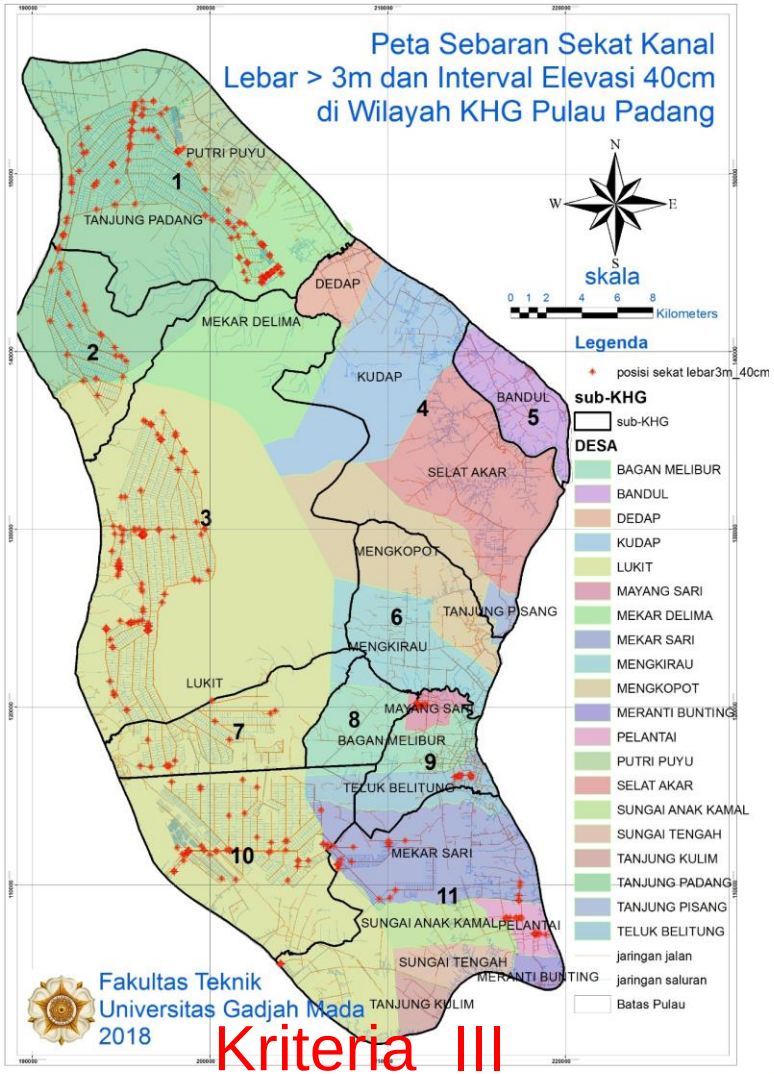
Skenario Titik Penyekatan Kanal



Kriteria I



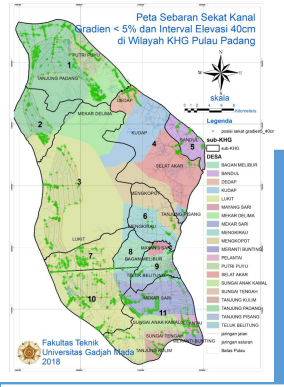
Kriteria II



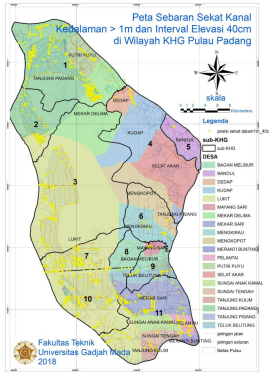
Kriteria III

Skenario Titik Penyekatan Kanal

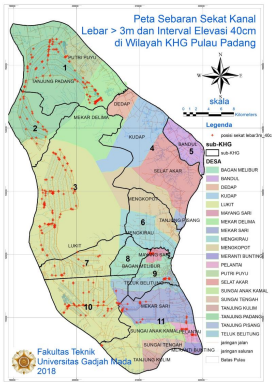
UNIVERSITAS GADJAH MADA



Kriteria 1:
perpotongan gradien kanal <5% & interval kontur setiap 40cm



Kriteria 2:
perpotongan dalam kanal >1m & interval kontur setiap 40cm



Kriteria 3:
perpotongan lebar kanal >3m & interval kontur setiap 40cm

SubKHG	Jumlah titik sekat kanal	SubKHG	Jumlah titik sekat kanal	SubKHG	Jumlah titik sekat kanal
1	802	1	599	1	167
2	116	2	86	2	21
3	383	3	349	3	207
4	148	4	24	4	0
5	68	5	6	5	0
6	70	6	52	6	0
7	186	7	154	7	13
8	6	8	5	8	0
9	92	9	68	9	36
10	318	10	259	10	53
11	416	11	304	11	52
total sekat	2605	total sekat	1906	total sekat	549

Hasil Perencanaan



Kesimpulan

- LIDAR dengan dukungan data lapangan dan data neraca air dapat dimanfaatkan untuk merencanakan pembasahan kembali.
- Kontribusi terpenting adalah analisis sebaran neraca dan kerawanana dapat diterapkan berdasarkan wilayah subKHG hasil identifikasi LIDAR dan desain sekat kanal berbasis informasi kedalaman dari lahan, lebar kanal dan gradien lahan dapat dengan mudah dilakukan menggunakan LIDAR.
- Apabila semua KHG rawan memiliki data detil maka metode yang sama dapat diterapkan untuk merencanakan pemabasahan kembali.

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Terima kasih

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